

RELATIVE ENTROPY DECAY VIA BKM COERCIVITY FOR QUANTUM MARKOV SEMIGROUPS

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ABSTRACT. We introduce the notion of the BKM coercivity constant and show that the positivity of this constant governs the exponential relative entropy decay of the semigroup. Indeed, we prove that the modified log-Sobolev constant is equivalent to the BKM coercivity constant up to a constant depending on the asymptotic conditional expectation. We also discover that the BKM coercivity constants for matrix amplifications are already attained with a qubit auxiliary system, which also coincides with the GNS spectral gap of the generating Lindbladian. As a corollary, we establish a criterion that the complete modified log-Sobolev inequality holds if and only if the GNS gap of the semigroup is strictly positive. All the discussion above applies to quantum Markov semigroups that admit a faithful asymptotic conditional expectation as long-time equilibration, with no detailed balance condition assumed. As an application, we show the finite-dimensional Chen-Kastoryano-Gilyén Gibbs samplers always satisfy the complete modified log-Sobolev inequality.

1. INTRODUCTION

Quantum Markov semigroups (QMS) describe the dissipative evolution of time-homogeneous open quantum systems. In recent decades, functional inequalities for quantum Markov semigroups have been intensively studied as tools for quantifying convergence to equilibrium [22, 16, 5, 1, 11, 13]. For a semigroup $P_t = e^{t\mathcal{L}}$ with asymptotic conditional expectation E , the modified logarithmic Sobolev inequality (MLSI) characterizes the exponential decay of relative entropy

$$D(P_{t*}\rho \| E_*\rho) \leq e^{-2\alpha t} D(\rho \| E_*\rho), \quad t \geq 0, \quad (1.1)$$

for some $\alpha > 0$ and every state ρ [27, 1]. Here D denotes quantum relative entropy. The maps P_{t*} and E_* are the trace adjoints of P_t and E , acting on states in the Schrödinger picture. Motivated by tensorization and applications to many-body systems, Gao, Junge, and LaRacuate introduced the complete modified logarithmic Sobolev inequality (CMLSI) [11]. It requires exponential decay (1.1) for every amplification $\text{id}_{M_n} \otimes P_t$, with a decay rate independent of n . This condition controls entropy decay even when the system is initially entangled with an ancillary system.

In finite dimensions, Gao and Rouzé proved positivity of the CMLSI constant under Gelfand–Naimark–Segal (GNS) symmetry, a quantum analog of the detailed balance condition [13]. Their estimate gives a quantitative lower bound in terms of the spectral gap and the complete Pimsner–Popa index. Later, using complete positivity order, Gao, Junge, LaRacuate, and Li obtained bounds with optimal logarithmic dependence on this index [12]. These comparisons provide a route from spectral gap to relative entropy decay under GNS symmetry.

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Davies semigroups are a natural setting for this approach. They arise as thermalization models in the weak-coupling limit [8], and their dissipative generators satisfy GNS symmetry. For one-dimensional commuting Hamiltonians, MLSI has been used to prove rapid thermalization [2]. Subsequent work established MLSI lower bounds that are uniform in system size [17]. Related results cover CSS models under a Dobrushin–Shlosman condition [29] and two-dimensional Abelian quantum double models [28]. These applications illustrate the effectiveness of entropy methods for commuting many-body systems.

For noncommuting Hamiltonians, a major breakthrough was the construction of an efficiently simulable Gibbs sampler by Chen, Kastoryano, and Gilyén [7]. Chen, Kastoryano, Brandão, and Gilyén developed this approach as a framework for quantum thermal simulation [6]. The resulting Lindblad dynamics preserve the exact Gibbs state through Kubo–Martin–Schwinger (KMS) symmetry, a weaker detailed balance condition than the GNS one.

The CKG construction replaces exact resolution of Bohr frequencies with Gaussian-filtered Heisenberg evolution. For local Hamiltonians and local input operators, Lieb–Robinson bounds ensure that the filtered jump operators are quasi-local, which is a crucial property that enables the efficient simulation on a quantum computer. The resulting generator is generally not GNS symmetric. This construction demonstrates the practical value of the weaker KMS condition: exact Gibbs stationarity can be combined with quasi-local dynamics and efficient quantum simulation [7, 6].

Given the efficient simulation, it still leaves the mixing time of the dynamics to be estimated. At high temperatures, Rouzé, França, and Alhambra first established polynomial-time Gibbs state preparation [25], and later they obtained optimal rapid mixing through exponential decay of an oscillator norm [26]. For one-dimensional finite-range Hamiltonians, Bergamaschi and Chen proved a positive lower bound on the KMS spectral gap at every fixed positive temperature [3]. The bound is uniform in system size and includes the Heisenberg chain. These results motivate the search for comparable relative entropy decay estimates beyond GNS symmetry.

Entropy methods in this setting face difficulties that are absent under GNS symmetry. Hamiltonian evolution can prevent CMLSI even in the presence of dissipation [18, Proposition 1.1]. For finite-dimensional semigroups with unital preduals, LaRacuente nevertheless proved complete entropy contraction toward the decoherence-free subspace at fixed positive time scales [18, Theorem 1.2]. These results do not provide a general MLSI or CMLSI criterion under KMS symmetry alone.

For the CKG sampler, efficient implementation and KMS spectral gap estimates do not by themselves establish relative entropy decay of the form (1.1). A central question is therefore whether MLSI and CMLSI can be established for these exact Gibbs dynamics beyond GNS symmetry. In this work, we obtain quantitative MLSI and CMLSI bounds for finite-dimensional QMS admitting a faithful asymptotic conditional expectation. Our criteria for positivity require no detailed balance. As an application, we prove that every fixed finite-dimensional CKG sampler has a positive CMLSI constant relative to its fixed-point algebra.

1.1. Main results. Throughout this subsection, let $P_t = e^{t\mathcal{L}}$ be a QMS on $B(\mathcal{H})$ with $\dim \mathcal{H} < \infty$. Assume that $\lim_{t \rightarrow \infty} P_t = E$ converges in norm to a faithful conditional expectation E . Write $\mathcal{N} = \text{Ran } E = \text{Fix}(P_t)$. We assume no detailed balance condition.

At a faithful invariant state ω , the Hessian of relative entropy defines the Bogoliubov–Kubo–Mori (BKM) metric

$$\langle A, A \rangle_{\text{BKM}, \omega} = \text{Tr}[A \Gamma_{\omega}^{-1}(A)]$$

where Γ_{ω}^{-1} is the inverse of

$$\Gamma_{\omega}(X) = \int_0^1 \omega^s X \omega^{1-s} ds.$$

The BKM coercivity constant is

$$\lambda_{\text{BKM}}(\mathcal{L}) := \inf_{\omega, A} \frac{-\text{Tr}[\mathcal{L}_*(A) \Gamma_{\omega}^{-1}(A)]}{\text{Tr}[A \Gamma_{\omega}^{-1}(A)]}.$$

The infimum ranges over faithful invariant states ω and nonzero self-adjoint A satisfying $E_* A = 0$ (see Definition 2.10). Let $\alpha(\mathcal{L})$ denote the optimal constant in (1.1). Recall that the Pimsner–Popa index is given by the optimal order constant

$$C(E) = \inf \{ C > 0 \mid \rho \leq C E_* \rho \text{ for all states } \rho \}.$$

Theorem 1.1 (MLSI and BKM coercivity). *Under the above assumptions,*

$$\frac{\lambda_{\text{BKM}}(\mathcal{L})}{2C(E)} \leq \alpha(\mathcal{L}) \leq \lambda_{\text{BKM}}(\mathcal{L}). \quad (1.2)$$

Thus MLSI holds exactly when $\lambda_{\text{BKM}}(\mathcal{L}) > 0$. The lower bound uses Frenkel’s integral formulas for relative entropy and its Hessian [10, Theorems 6 and 7(a)]. Under GNS symmetry, $\lambda_{\text{BKM}}(\mathcal{L})$ equals the usual spectral gap of $-\mathcal{L}$. The theorem therefore recovers the ordinary index bound of [13, Theorem 3.3] in that case.

We next adjoin an ancillary system. For $n \geq 1$, write

$$\mathcal{L}^{(n)} := \text{id}_{M_n} \otimes \mathcal{L}, \quad E^{(n)} := \text{id}_{M_n} \otimes E.$$

We define the complete BKM coercivity constant as

$$\lambda_{\text{BKM}}^c(\mathcal{L}) := \inf_{n \geq 1} \lambda_{\text{BKM}}(\mathcal{L}^{(n)}).$$

We show that the complete BKM coercivity constant coincides with the GNS spectral gap. For a general QMS, this gap is defined without a symmetry assumption by

$$\lambda_{\text{GNS}}(\mathcal{L}) := \inf_{\substack{X \neq 0 \\ E(X)=0}} \frac{-\Re \text{Tr}(\omega X^* \mathcal{L}(X))}{\text{Tr}(\omega X^* X)}.$$

Theorem 1.2 (BKM coercivity under amplification). *Under the above assumptions,*

$$\lambda_{\text{BKM}}^c(\mathcal{L}) = \lambda_{\text{BKM}}(\text{id}_{M_2} \otimes \mathcal{L}) = \lambda_{\text{GNS}}(\mathcal{L}) \leq \lambda_{\text{BKM}}(\mathcal{L}). \quad (1.3)$$

Without detailed balance, Wirth compared the GNS gap with gaps associated with operator-monotone functions [32, Corollary 3.7 and Remark 3.9]. His comparison includes the BKM metric and gives the lower bound by the GNS gap. Our result establishes equality after amplification by M_2 .

We now define the complete MLSI constants by

$$\alpha^c(\mathcal{L}) := \inf_{n \geq 1} \alpha(\mathcal{L}^{(n)}).$$

The complete Pimsner–Popa index is $C_{\text{cb}}(E) := \sup_{n \geq 1} C(E^{(n)})$. Combining Theorem 1.2 with Theorem 1.1 gives the complete estimate.

Corollary 1.3 (CMLSI criterion). *Under the above assumptions,*

$$\frac{\lambda_{\text{GNS}}(\mathcal{L})}{2C_{\text{cb}}(E)} \leq \alpha^c(\mathcal{L}) \leq \lambda_{\text{GNS}}(\mathcal{L}). \quad (1.4)$$

Consequently,

$$\alpha^c(\mathcal{L}) > 0 \iff \lambda_{\text{GNS}}(\mathcal{L}) > 0. \quad (1.5)$$

The corollary follows by applying Theorem 1.1 at every matrix level and using $C(E^{(n)}) \leq C_{\text{cb}}(E)$. We note that under GNS symmetry, both BKM constants and the GNS and KMS gaps coincide with the spectral gap of $-\mathcal{L}$ (see Corollary 5.7). The complete estimate above then recovers [13] with no detailed balance being assumed.

For CKG samplers, the GNS Dirichlet form is an integral of squared commutator norms involving the filtered jump operators. We show that its kernel is exactly the fixed-point algebra, which in finite dimensions implies CMLSI (see Section 7).

The paper is organized as follows. Section 2 introduces the notation and basic assumptions. Section 3 proves the comparison between BKM coercivity and MLSI. Section 4 characterizes the ordinary BKM constant through endpoint and internal constants. Section 5 proves the amplification theorem and treats the GNS-symmetric case. Section 6 gives the examples separating the spectral gaps and entropy decay constants. Section 7 applies the CMLSI criterion to CKG samplers.

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2. PRELIMINARIES

2.1. Quantum Markov semigroups and detailed balance. Throughout, $\mathcal{H}$ is finite-dimensional and Tr denotes the unnormalized trace.

Definition 2.1. A quantum Markov semigroup (QMS) $(P_t)_{t \geq 0}$ on $B(\mathcal{H})$ is a norm-continuous semigroup of unital completely positive maps satisfying $P_0 = \text{id}$ and $P_s \circ P_t = P_{s+t}$ for all $s, t \geq 0$.

The generator of the semigroup is defined

$$\mathcal{L}(X) := \lim_{t \rightarrow 0} \frac{1}{t} (P_t(X) - X).$$

Then $P_t = e^{t\mathcal{L}}$, and the generator admits the Gorini–Kossakowski–Lindblad–Sudarshan (GKLS) form [15, 21]

$$\mathcal{L}(X) = i[H, X] + \sum_j \left(V_j^* X V_j - \frac{1}{2} \{V_j^* V_j, X\} \right), \quad (2.1)$$

with $H = H^*$ and $V_j \in B(\mathcal{H})$. Its trace predual $P_{t*} = e^{t\mathcal{L}^*}$ is given by

$$\text{Tr}(X P_{t*}(\rho)) = \text{Tr}(P_t(X) \rho), \forall X, \rho \in B(\mathcal{H}).$$

A state σ is invariant if $P_{t*}(\sigma) = \sigma$ for every $t \geq 0$, or equivalently $\mathcal{L}_*(\sigma) = 0$. For a faithful invariant state $\sigma > 0$, the GNS and KMS inner products are

$$\langle X, Y \rangle_{\text{GNS}, \sigma} := \text{Tr}(\sigma X^* Y), \quad \langle X, Y \rangle_{\text{KMS}, \sigma} := \text{Tr}(\sigma^{1/2} X^* \sigma^{1/2} Y). \quad (2.2)$$

The QMS is called *GNS symmetric* (respectively, *KMS symmetric*) with respect to σ if $\mathcal{L}$ is self-adjoint for the corresponding inner product. These conditions are also called *GNS-* and *KMS-detailed balance*. GNS symmetry implies KMS symmetry, whereas the converse need not hold [5, Theorem 2.9 and Appendix B]. Indeed, It is proved that a semigroup P_t is GNS-symmetry if and only if P_t is KMS-symmetry and covariant under modular automorphism group

$$\sigma^{is} P_t(\sigma^{-is} \cdot \sigma^{is}) \sigma^{-is} = P_t(\cdot), \forall t \geq 0.$$

Under GNS and KMS symmetry with respect to a faithful invariant state σ , respectively, define the corresponding spectral gaps [32, Section 3] by

$$\lambda_{\text{GNS}}(\mathcal{L}) := \inf_X \frac{-\langle X, \mathcal{L}(X) \rangle_{\text{GNS}, \sigma}}{\langle X, X \rangle_{\text{GNS}, \sigma}}, \quad \lambda_{\text{KMS}}(\mathcal{L}) := \inf_X \frac{-\langle X, \mathcal{L}(X) \rangle_{\text{KMS}, \sigma}}{\langle X, X \rangle_{\text{KMS}, \sigma}}.$$

Here each infimum is over nonzero X orthogonal to $\text{Ker } \mathcal{L}$ in the corresponding inner product. In either case, the gap equals $\min(\text{spec}(-\mathcal{L}) \setminus \{0\})$ and has the same value for every faithful invariant reference state. See [13, Lemma 3.2] for the GNS case.

2.2. Conditional expectations and indices.

Definition 2.2. Let $\mathcal{N} \subseteq \text{B}(\mathcal{H})$ be a unital $*$ -subalgebra. A conditional expectation $E : \text{B}(\mathcal{H}) \rightarrow \mathcal{N}$ is a unital completely positive map onto $\mathcal{N}$ satisfying $E \circ E = E$. It is called faithful if $E(X^*X) = 0$ implies $X = 0$. Its trace adjoint is denoted by E_* .

It is proved [31, Theorem 1] that the conditional expectation satisfies the bimodule property

$$E(aXb) = aE(X)b, \quad a, b \in \mathcal{N}, \quad X \in \text{B}(\mathcal{H}). \quad (2.3)$$

Throughout the paper, we will work under the following assumption.

Assumption 2.3. We assume that the limit

$$E := \lim_{t \rightarrow \infty} P_t \quad (2.4)$$

exists and is a faithful conditional expectation.

Given the assumption, set the fixed-point subspace

$$\text{Fix}(P) := \{X \in \text{B}(\mathcal{H}) : P_t(X) = X \text{ for every } t \geq 0\}, \quad \mathcal{N} := \text{Ran } E = \text{Fix}(P). \quad (2.5)$$

In particular,

$$P_t E = E P_t = E, \quad P_{t*} E_* = E_* P_{t*} = E_*. \quad (2.6)$$

Since E is unital and faithful and $P_{t*} E_* = E_*$, the state $E_*(\rho)$ is faithful and invariant whenever ρ is faithful. In particular, a canonical choice $\sigma_0 = E_*(I/d)$, where $d = \dim \mathcal{H}$.

Under GNS or KMS symmetry, E is the orthogonal projection onto $\text{Ker } \mathcal{L}$ for the corresponding inner product. The orthogonality conditions defining the gaps in Section 2.1 can therefore be written as $E(X) = 0$.

The QMS is called *primitive* if the fixed space $\mathcal{N} = \mathbb{C}I$ is trivial. In this case, $E(X) = \text{Tr}(\sigma X)I$ is the depolarizing map for the unique faithful invariant state σ .

We recall the standard form of a faithful conditional expectation [13, Section 2.1, Eq. (13)].

Proposition 2.4 (Structure of conditional expectations). *Let $\mathcal{H}$ be finite-dimensional, $\mathcal{N} \subseteq \mathcal{B}(\mathcal{H})$ be a unital $*$ -subalgebra, and let $E : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{N}$ be a faithful conditional expectation. Up to unitary equivalence, there exist a finite index set $\mathcal{I}$ and finite-dimensional Hilbert spaces $\mathcal{K}_k, \mathcal{R}_k, k \in \mathcal{I}$ such that*

$$\mathcal{H} = \bigoplus_{k \in \mathcal{I}} \mathcal{K}_k \otimes \mathcal{R}_k, \quad \mathcal{N} = \bigoplus_{k \in \mathcal{I}} \mathcal{B}(\mathcal{K}_k) \otimes I_{\mathcal{R}_k}. \quad (2.7)$$

Let p_k denote the projection onto the k th summand, and write $X_{kk} := p_k X p_k$ and $\rho_{kk} := p_k \rho p_k$. There are faithful density matrices $\sigma_k \in \mathcal{B}(\mathcal{R}_k)$ for which

$$E(X) = \bigoplus_{k \in \mathcal{I}} [(\text{id}_{\mathcal{K}_k} \otimes \varphi_{\sigma_k})(X_{kk})] \otimes I_{\mathcal{R}_k}, \quad (2.8)$$

$$E_*(\rho) = \bigoplus_{k \in \mathcal{I}} \text{Tr}_{\mathcal{R}_k}(\rho_{kk}) \otimes \sigma_k, \quad (2.9)$$

where $\varphi_{\sigma_k}(Z) := \text{Tr}(\sigma_k Z)$. The states fixed by E_* are precisely those of the form

$$\rho = \bigoplus_{k \in \mathcal{I}} \eta_k \otimes \sigma_k, \quad \eta_k \geq 0, \quad \sum_{k \in \mathcal{I}} \text{Tr} \eta_k = 1, \quad (2.10)$$

where $\eta_k \in \mathcal{B}(\mathcal{K}_k)$. Such an invariant state ρ is faithful if and only if η_k is strictly positive for every k .

Let $\mathcal{D}(\mathcal{H})$ denote the set of states (density operators) on $\mathcal{H}$, and set the set of invariant states as

$$\mathcal{D}(E) := \{\sigma \in \mathcal{D}(\mathcal{H}) : E_*(\sigma) = \sigma\}. \quad (2.11)$$

Definition 2.5. The ordinary and complete Pimsner–Popa indices [24, 13] of a faithful conditional expectation E are defined by

$$C(E) := \inf\{C > 0 : \rho \leq CE_*\rho \text{ for every } \rho \geq 0\}, \quad (2.12)$$

$$C_{\text{cb}}(E) := \sup_{n \geq 1} C(\text{id}_{M_n} \otimes E). \quad (2.13)$$

We have the following formula for Pimsner–Popa indices in finite dimensions.

Proposition 2.6 ([14, Proposition 3.2]). *Let E be a faithful conditional expectation in the form of Proposition 2.4. Let $m_k = \dim \mathcal{K}_k$ and $d_k = \dim \mathcal{R}_k$, and list the eigenvalues of σ_k^{-1} in decreasing order as $\lambda_1^\downarrow(\sigma_k^{-1}), \dots, \lambda_{d_k}^\downarrow(\sigma_k^{-1})$, counted with multiplicity. Then*

$$C(E) = \sum_{k \in \mathcal{I}} \sum_{j=1}^{\min\{m_k, d_k\}} \lambda_j^\downarrow(\sigma_k^{-1}), \quad C_{\text{cb}}(E) = \sum_{k \in \mathcal{I}} \text{Tr}(\sigma_k^{-1}). \quad (2.14)$$

Via trace duality, the Pimsner–Popa indices are equivalently formulated in the Heisenberg picture

$$C(E) = \inf\{C > 0 : X \leq CE(X) \text{ for all } X \geq 0\}, \quad (2.15)$$

$$C_{\text{cb}}(E) = \inf\{C > 0 : CE - \text{id} \text{ is completely positive}\}. \quad (2.16)$$

Example 2.7. The ordinary and complete indices need not coincide. For $\mathcal{H} = \mathcal{K} \otimes \mathbb{C}^d$, consider the conditional expectation $E(X) = [(\text{id}_{\mathcal{B}(\mathcal{K})} \otimes \frac{\text{Tr}}{d})(X)] \otimes I_d$. Then

$$C(E) = d \min\{\dim \mathcal{K}, d\}, \quad C_{\text{cb}}(E) = d^2.$$

For the replacer channel $E_*(\rho) = \text{Tr}(\rho)\sigma$, where σ is a faithful state, the indices are

$$C(E) = \|\sigma^{-1}\|_{\infty}, \quad C_{\text{cb}}(E) = \text{Tr}(\sigma^{-1}).$$

2.3. Modified log-Sobolev constant. For states ρ and σ , define the Umegaki relative entropy

$$D(\rho\|\sigma) := \text{Tr} \rho(\log \rho - \log \sigma), \quad (2.17)$$

given the support condition $\text{supp}(\rho) \subset \text{supp}(\sigma)$. The definition extends to the pair ρ and σ with equal trace. Relative to the conditional expectation E , define

$$D_E(\rho) := \inf_{\sigma \in \mathcal{D}(E)} D(\rho\|\sigma) = D(\rho\|E_*\rho). \quad (2.18)$$

Here the infimum in (2.18) is attained at $E_*\rho$ by the chain rule [13, Section 2.1, Eq. (12)] is

$$D(\rho\|\omega) = D(\rho\|E_*\rho) + D(E_*\rho\|\omega), \quad \omega \in \mathcal{D}(E), \quad (2.19)$$

In particular, $D(\rho\|E_*\rho) < \infty$ because $\rho \leq C(E)E_*\rho$.

For a faithful state ρ , define the entropy production [27]

$$\text{EP}_{\mathcal{L}}(\rho) := -\text{Tr}[\mathcal{L}_*(\rho)(\log \rho - \log E_*\rho)] = -\left.\frac{d}{dt}\right|_{t=0^+} D(P_{t*}\rho\|E_*\rho). \quad (2.20)$$

The entropy production $\text{EP}_{\mathcal{L}}$ is always nonnegative by data processing inequality, since $P_{t*}E_* = E_*$.

Definition 2.8. The modified log-Sobolev inequality (MLSI) constant of $\mathcal{L}$ relative to $\mathcal{N}$ is

$$\alpha(\mathcal{L}) := \inf_{\substack{\rho \in \mathcal{D}(\mathcal{H}), \rho > 0 \\ \rho \neq E_*\rho}} \frac{\text{EP}_{\mathcal{L}}(\rho)}{2D(\rho\|E_*\rho)}. \quad (2.21)$$

Equivalently, $\alpha(\mathcal{L}) \geq \alpha$ means

$$\text{EP}_{\mathcal{L}}(\rho) \geq 2\alpha D(\rho\|E_*\rho) \quad (2.22)$$

for every faithful state. By standard Gronwall Lemma, (2.22) is equivalent to

$$D(P_{t*}\rho\|E_*\rho) \leq e^{-2\alpha t} D(\rho\|E_*\rho), \quad t \geq 0, \quad (2.23)$$

for every faithful state ρ . For $\alpha > 0$, (2.23) implies $P_t \rightarrow E$. For MLSI in the non-primitive setting was introduced in Bardet [1] and studied in [13, 12]. We will use the normalization $\text{EP} \geq 2\alpha D$ throughout.

For $n \geq 1$, set the matrix level amplification

$$\mathcal{L}^{(n)} := \text{id}_{M_n} \otimes \mathcal{L}, \quad E^{(n)} := \text{id}_{M_n} \otimes E,$$

and let $\alpha(\mathcal{L}^{(n)})$ denote the corresponding MLSI constant of $\mathcal{L}^{(n)}$ relative to $\text{Ran } E^{(n)} = M_n \otimes \mathcal{N}$. The complete MLSI constant is defined as

$$\alpha^c(\mathcal{L}) := \inf_{n \geq 1} \alpha(\mathcal{L}^{(n)}). \quad (2.24)$$

Namely, $\alpha^c(\mathcal{L}) \geq \alpha$ if and only if (2.23) holds for every finite matrix amplification. The complete MLSI constant was introduced in [11] for the tensorization property that for two quantum Markov semigroups $P_t = e^{\mathcal{L}_1 t}$ and $Q_t = e^{\mathcal{L}_2 t}$,

$$\alpha^c(\mathcal{L}_1 \otimes \text{id} + \text{id} \otimes \mathcal{L}_2) = \min\{\alpha^c(\mathcal{L}_1), \alpha^c(\mathcal{L}_2)\}.$$

2.4. BKM metric and coercivity constants. For $\sigma > 0$, define the Bogoliubov–Kubo–Mori (BKM) map [23, 19]

$$\Gamma_\sigma(X) := \int_0^1 \sigma^s X \sigma^{1-s} ds. \quad (2.25)$$

Given the spectrum decomposition $\sigma = \sum_i \lambda_i |i\rangle\langle i|$ and $E_{ij} := |i\rangle\langle j|$, then

$$\Gamma_\sigma(E_{ij}) = m(\lambda_i, \lambda_j) E_{ij}, \quad m(a, b) = \frac{a - b}{\log a - \log b} \quad (2.26)$$

with $m(a, a) = a$ by continuity. The inverse of Γ_σ defines the Hessian metric of relative entropy.

Lemma 2.9. *For every $\rho > 0$ and $X = X^* \in \mathcal{B}(\mathcal{H})$, the Fréchet derivative of the matrix logarithm satisfies [20, Section III]*

$$\begin{aligned} D(\log)_\rho[X] &:= \left. \frac{d}{du} \right|_{u=0} \log(\rho + uX) \\ &= \int_0^\infty (\rho + sI)^{-1} X (\rho + sI)^{-1} ds = \Gamma_\rho^{-1}(X). \end{aligned} \quad (2.27)$$

For strictly positive states ρ, σ , set $A := \rho - \sigma$ and $\rho_t := t\rho + (1-t)\sigma$. Consequently,

$$\log \rho - \log \sigma = \int_0^1 \Gamma_{\rho_s}^{-1}(A) ds. \quad (2.28)$$

Moreover, the relative entropy admits the integral representation [13, Lemma 2.2]

$$D(\rho \| \sigma) = \int_0^1 (1-t) \text{Tr}[A \Gamma_{\rho_t}^{-1}(A)] dt. \quad (2.29)$$

In particular, for every strictly positive state σ and traceless self-adjoint A ,

$$D(\sigma + \varepsilon A \| \sigma) = \frac{\varepsilon^2}{2} \text{Tr}[A \Gamma_\sigma^{-1}(A)] + o(\varepsilon^2) \quad (\varepsilon \rightarrow 0). \quad (2.30)$$

Let σ be a faithful invariant state and let $A = A^*$ satisfy $E_* A = 0$. Since E_* is trace preserving, $\text{Tr} A = 0$. Thus, for all sufficiently small real ε , $\sigma + \varepsilon A$ is a faithful state with $E_*(\sigma + \varepsilon A) = \sigma$. Expanding (2.20) gives

$$\text{EP}_\mathcal{L}(\sigma + \varepsilon A) = -\varepsilon^2 \text{Tr}[\mathcal{L}_*(A) \Gamma_\sigma^{-1}(A)] + o(\varepsilon^2). \quad (2.31)$$

Definition 2.10. The *BKM coercivity constant* of $\mathcal{L}$ is

$$\lambda_{\text{BKM}}(\mathcal{L}) := \inf_{\substack{\sigma > 0, \text{Tr } \sigma = 1, \mathcal{L}_*(\sigma) = 0 \\ A = A^* \neq 0, E_* A = 0}} \frac{-\text{Tr}[\mathcal{L}_*(A)\Gamma_\sigma^{-1}(A)]}{\text{Tr}[A\Gamma_\sigma^{-1}(A)]}, \quad (2.32)$$

The infimum ranges over both faithful invariant states σ and centered self-adjoint directions A . For $n \geq 1$, write $\lambda_{\text{BKM}}^{(n)}(\mathcal{L}) := \lambda_{\text{BKM}}(\text{id}_{M_n} \otimes \mathcal{L})$. The *complete BKM coercivity constant* is

$$\lambda_{\text{BKM}}^c(\mathcal{L}) := \inf_{n \geq 1} \lambda_{\text{BKM}}^{(n)}(\mathcal{L}). \quad (2.33)$$

Linearization gives the following upper bounds.

Proposition 2.11. *Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Then*

$$\alpha(\mathcal{L}) \leq \lambda_{\text{BKM}}(\mathcal{L}), \quad \alpha^c(\mathcal{L}) \leq \lambda_{\text{BKM}}^c(\mathcal{L}). \quad (2.34)$$

Proof. Fix an admissible pair (σ, A) in the infimum of Definition 2.10. For every sufficiently small nonzero real ε , the state $\rho_\varepsilon := \sigma + \varepsilon A$ is faithful and $E_* \rho_\varepsilon = \sigma$. By (2.30) and (2.31),

$$\lim_{\varepsilon \rightarrow 0} \frac{\text{EP}_{\mathcal{L}}(\rho_\varepsilon)}{2D(\rho_\varepsilon \| E_* \rho_\varepsilon)} = \frac{-\text{Tr}[\mathcal{L}_*(A)\Gamma_\sigma^{-1}(A)]}{\text{Tr}[A\Gamma_\sigma^{-1}(A)]}.$$

Taking the infimum over admissible (σ, A) proves the bound. Applying the same argument to $\text{id}_{M_n} \otimes \mathcal{L}$ for every n , and then taking the infimum over n , proves the complete bound. $\square$

3. MLSI BOUND VIA BKM COERCIVITY

We derive lower bounds for the MLSI constants in terms of BKM coercivity and the Pimsner–Popa indices. The proof uses Frenkel’s formula to express the relative entropy and BKM metric contraction deficits as integrals against the same positive measure.

For $\sigma > 0$ and $A = A^*$, write

$$H_\sigma(A) := \text{Tr}[A\Gamma_\sigma^{-1}(A)], \quad Q_{\sigma, \mathcal{L}}(A) := -\text{Tr}[\mathcal{L}_*(A)\Gamma_\sigma^{-1}(A)], \quad (3.1)$$

and abbreviate $Q_{\sigma, \mathcal{L}}$ to Q_σ whenever the generator is clear.

3.1. An entropy-metric comparison. For states ρ, σ with $\sigma > 0$, $H_\sigma(\rho - \sigma)$ is the quantum χ^2 -divergence associated with BKM metric; see [30, Definition 1 and Eq. (54)]. We recall the entropy-metric comparison from [13, Lemma 2.2] (see also [19, 30]).

Lemma 3.1. *Let $\rho \geq 0$ and $\sigma > 0$ have equal trace and $\rho \leq C\sigma$. Then*

$$D(\rho \| \sigma) \leq H_\sigma(\rho - \sigma). \quad (3.2)$$

We present a comparison lemma for the contraction difference in terms of relative entropy and BKM metric. It is an application of the Frenkel’s integral formula of relative entropy [10, Theorem 6]

$$D(\rho \| \sigma) = \int_{-\infty}^{\infty} \frac{\text{Tr}^-((1-u)\rho + u\sigma)}{|u|(u-1)^2} du. \quad (3.3)$$

Here, for a Hermitian operator X , we denote

$$\text{Tr}^-(X) := \text{Tr}(X_-), \quad X_- := \frac{|X| - X}{2}.$$

Lemma 3.2. *Let $\rho \geq 0$ and $\sigma > 0$ have equal trace, and let T_* be a positive trace-preserving map satisfying $T_*(\sigma) = \sigma$. If $\rho \leq C\sigma$ for some $C > 1$, then*

$$D(\rho\|\sigma) - D(T_*\rho\|\sigma) \geq \frac{1}{2C} [H_\sigma(\rho - \sigma) - H_\sigma(T_*\rho - T_*\sigma)]. \quad (3.4)$$

Proof. Set $A = \rho - \sigma$. The assumptions give

$$\text{Tr } A = 0, \quad -\sigma \leq A \leq (C - 1)\sigma.$$

For every real r for which $\rho_r := \sigma + rA \geq 0$, define

$$\Delta_T(r) := D(\rho_r\|\sigma) - D(T_*\rho_r\|\sigma). \quad (3.5)$$

We also set

$$\delta_T(z) := \text{Tr}^-(\sigma + zA) - \text{Tr}^-(T_*(\sigma + zA)), \quad z \in \mathbb{R}. \quad (3.6)$$

The quantity $\delta_T(z)$ is the decrease in the trace of the negative part of $\sigma + zA$ under T_* , which is always non-negative by the monotonicity $\text{Tr}^-(T_*X) \leq \text{Tr}^-(X)$ for positive and trace preserving T_* and self-adjoint $X^* = X$.

Step 1: the defect measure. For $0 \leq z \leq 1$,

$$\sigma + zA = (1 - z)\sigma + z\rho \geq 0.$$

For $-1/(C - 1) \leq z \leq 0$, the upper bound $A \leq (C - 1)\sigma$ gives

$$\sigma + zA \geq (1 + z(C - 1))\sigma \geq 0.$$

Positivity of T_* gives $T_*(\sigma + zA) \geq 0$ throughout $[-1/(C - 1), 1]$. Consequently,

$$\delta_T(z) = 0 \quad \text{for} \quad -\frac{1}{C - 1} \leq z \leq 1. \quad (3.7)$$

Moreover, $X \mapsto \text{Tr}^-(X)$ is trace-norm continuous and

$$0 \leq \delta_T(z) \leq \text{Tr}^-(\sigma + zA) \leq \|\sigma\|_1 + |z| \|A\|_1.$$

Define

$$d\nu_T(z) := \frac{\delta_T(z)}{|z|^3} dz, \quad (3.8)$$

where the density is set to zero at $z = 0$. Note that the Eq. (3.7) removes the singularity at the origin, while the linear growth bound makes the density $O(|z|^{-2})$ at infinity. Thus ν_T is a finite positive measure.

Step 2: The entropy difference. Apply (3.3) to $\rho_r = \sigma + rA$ and $T_*\rho_r = \sigma + rT_*A$. Since $\sigma > 0$, both relative entropies and both Frenkel integrals are finite. Set $z = r(1 - u)$. Using $T_*(\sigma) = \sigma$, one has

$$\begin{aligned} (1 - u)\rho_r + u\sigma &= \sigma + zA, \\ (1 - u)T_*\rho_r + u\sigma &= T_*(\sigma + zA), \\ |u|(u - 1)^2 &= \frac{|z - r|z^2}{|r|^3}, \quad |du| = \frac{|dz|}{|r|}. \end{aligned}$$

Taking the absolute value of the Jacobian gives, for either sign of r ,

$$\Delta_T(r) = D(\rho_r\|\sigma) - D(T_*\rho_r\|\sigma) = r^2 \int_{-\infty}^{\infty} \frac{\delta_T(z)}{z^2|z - r|} dz. \quad (3.9)$$

To check local integrability at $z = r$, use $\rho_r \geq 0$ and the variational formula for Tr^- to obtain

$$\text{Tr}^-(\sigma + zA) \leq \|(\sigma + zA) - \rho_r\|_1 = |z - r| \|A\|_1.$$

Equivalently, using the measure ν_T , we have

$$\frac{\Delta_T(r)}{r^2} = \int_{\mathbb{R}} \frac{|z|}{|z - r|} d\nu_T(z). \quad (3.10)$$

Step 3: The total mass of ν_T . Set

$$\eta := \min \left\{ 1, \frac{1}{C - 1} \right\} > 0.$$

By (3.7), $\nu_T([- \eta, \eta]) = 0$. Hence, for $|r| \leq \eta/2$,

$$\frac{|z|}{|z - r|} \leq 2 \quad \text{for } \nu_T\text{-almost every } z.$$

The kernel converges to 1 for ν_T -almost every z as $r \rightarrow 0$. Dominated convergence in (3.10) therefore gives

$$\lim_{r \rightarrow 0} \frac{\Delta_T(r)}{r^2} = \nu_T(\mathbb{R}). \quad (3.11)$$

A positive map preserves Hermitian operators, and trace preservation gives $\text{Tr}(T_*A) = 0$. For all sufficiently small real r , both $\sigma + rA$ and $\sigma + rT_*A$ are positive definite. By homogeneity, the BKM expansion (2.30) gives

$$\Delta_T(r) = \frac{r^2}{2} [H_\sigma(A) - H_\sigma(T_*A)] + o(r^2).$$

Comparing with (3.11), we obtain the exact identity

$$\nu_T(\mathbb{R}) = \frac{1}{2} [H_\sigma(A) - H_\sigma(T_*A)]. \quad (3.12)$$

Thus the BKM contraction deficit is twice the total mass of ν_T . We note that the same identity essentially follows by subtracting Frenkel's Hessian formulas [10, Theorem 7(a), $m = 2$] for A and T_*A at σ .

Step 4: Compare the two kernels. Taking $r = 1$ in (3.10) gives

$$D(\rho \|\sigma) - D(T_*\rho \|\sigma) = \int_{\mathbb{R}} \frac{|z|}{|z - 1|} d\nu_T(z). \quad (3.13)$$

By (3.7), ν_T is concentrated on

$$\left(-\infty, -\frac{1}{C - 1} \right) \cup (1, \infty).$$

The claim then follows from the elementary inequality

$$\frac{|z|}{|z - 1|} \geq \frac{1}{C}, \quad z > 1 \text{ or } z < -\frac{1}{C - 1}. \quad \square$$

3.2. From tangent coercivity to entropy decay. We now apply the above lemma to $T_* = P_{s*}$ and differentiate at $s = 0$.

Theorem 3.3. *Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Then every faithful state $\rho \in \mathcal{D}(\mathcal{H})$ satisfies*

$$\text{EP}_{\mathcal{L}}(\rho) \geq \frac{1}{C(E)} Q_{E_*\rho}(\rho - E_*\rho). \quad (3.14)$$

Consequently,

$$\alpha(\mathcal{L}) \geq \frac{\lambda_{\text{BKM}}(\mathcal{L})}{2C(E)}. \quad (3.15)$$

Proof. Fix a faithful state ρ and set $\sigma = E_*\rho$, $A = \rho - \sigma$, and $C = C(E)$. The state σ is faithful and invariant. The index bound gives $\rho \leq C\sigma$. For $s > 0$, applying Lemma 3.2 to $T_* = P_{s*}$ gives

$$\frac{D(\rho\|\sigma) - D(P_{s*}\rho\|\sigma)}{s} \geq \frac{1}{2C} \frac{H_\sigma(A) - H_\sigma(P_{s*}A)}{s}. \quad (3.16)$$

As $s \downarrow 0$, the left-hand side converges to $\text{EP}_{\mathcal{L}}(\rho)$, whereas self-adjointness of Γ_σ^{-1} for the Hilbert–Schmidt inner product gives

$$-\frac{1}{2} \frac{d}{ds} H_\sigma(P_{s*}A) \Big|_{s=0} = -\text{Tr}[\mathcal{L}_*(A)\Gamma_\sigma^{-1}(A)] = Q_\sigma(A). \quad (3.17)$$

Hence $\text{EP}_{\mathcal{L}}(\rho) \geq C^{-1}Q_\sigma(A)$. Lemma 3.1 and definition of $\lambda_{\text{BKM}}(\mathcal{L})$ then yield

$$\text{EP}_{\mathcal{L}}(\rho) \geq \frac{1}{C} Q_\sigma(A) \geq \frac{\lambda_{\text{BKM}}(\mathcal{L})}{C} H_\sigma(A) \geq \frac{\lambda_{\text{BKM}}(\mathcal{L})}{C} D(\rho\|\sigma),$$

which is (3.15) in the normalization $\text{EP} \geq 2\alpha D$. $\square$

Applying the above theorem to $\mathcal{L}^{(n)} = \text{id}_{M_n} \otimes \mathcal{L}$ with $C = C(\text{id}_{M_n} \otimes E)$ for every $n \geq 1$ gives the complete MLSI bound.

Corollary 3.4. *Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Then*

$$\frac{\lambda_{\text{BKM}}(\mathcal{L})}{2C(E)} \leq \alpha(\mathcal{L}) \leq \lambda_{\text{BKM}}(\mathcal{L}), \quad (3.18)$$

and

$$\frac{\lambda_{\text{BKM}}^c(\mathcal{L})}{2C_{\text{cb}}(E)} \leq \alpha^c(\mathcal{L}) \leq \lambda_{\text{BKM}}^c(\mathcal{L}). \quad (3.19)$$

Hence positivity of the BKM coercivity constant is equivalent to positivity of the corresponding MLSI constant, for both the ordinary and complete constants.

The above corollary extends [13, Theorem 3.3] beyond GNS symmetry. Indeed, we will show in Corollary 5.7 that λ_{BKM} coincides with both the KMS and GNS spectral gaps under GNS symmetry.

4. CORNER DECOMPOSITION AND BKM COERCIVITY

In this section, we restrict the generator to the rectangular corners of the fixed-point decomposition and express the BKM forms in these coordinates. Endpoint interpolation then yields an exact expression for the ordinary BKM coercivity constant. We use the notation H_ω and Q_ω from (3.1).

4.1. Fixed-point structure and corner generators. Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Set $\mathcal{N} = \text{Ran } E$. Every $a \in \mathcal{N}$ belongs to the multiplicative domain of P_t , so P_t is $\mathcal{N}$ -bimodular. Differentiation gives the same property for $\mathcal{L}$. Thus, for every $a, b \in \mathcal{N}$, $X \in \text{B}(\mathcal{H})$, and $t \geq 0$,

$$P_t(aXb) = aP_t(X)b, \quad \mathcal{L}(aXb) = a\mathcal{L}(X)b. \quad (4.1)$$

Such an expectation exists whenever P_t is KMS symmetric with respect to a faithful invariant state σ . Indeed, $\mathcal{L}$ is diagonalizable with spectrum in $(-\infty, 0]$, so $P_t = e^{t\mathcal{L}}$ converges to the projection onto $\text{Ker } \mathcal{L}$.

By the orthogonal decomposition in Proposition 2.4, we may write

$$\mathcal{H} = \bigoplus_{k \in \mathcal{I}} \mathcal{K}_k \otimes \mathcal{R}_k, \quad \mathcal{N} = \bigoplus_{k \in \mathcal{I}} \text{B}(\mathcal{K}_k) \otimes I_{\mathcal{R}_k}, \quad \omega = \bigoplus_{k \in \mathcal{I}} \eta_k \otimes \sigma_k. \quad (4.2)$$

Here the faithful density matrices σ_k are determined by E , while a faithful invariant state ω is specified by

$$\eta_k \in \text{B}(\mathcal{K}_k), \quad \eta_k > 0, \quad \sum_k \text{Tr } \eta_k = 1.$$

Let p_k be the projection onto $\mathcal{K}_k \otimes \mathcal{R}_k$, and set

$$\mathbb{C}_{kl} := p_k \text{B}(\mathcal{H}) p_l = \text{B}(\mathcal{K}_l, \mathcal{K}_k) \otimes \text{B}(\mathcal{R}_l, \mathcal{R}_k).$$

Since $p_k, p_l \in \mathcal{N}$, the bimodularity makes each $\mathbb{C}_{kl}$ invariant under $\mathcal{L}$. Fix orthonormal bases of the noiseless factors $\text{B}(\mathcal{K}_l, \mathcal{K}_k)$ and write the matrix entry as

$$E_{ij}^{kl} := |k, i\rangle \langle l, j| \in \text{B}(\mathcal{K}_l, \mathcal{K}_k), \quad (E_{ij}^{kl})^* = E_{ji}^{lk}.$$

For each (k, l) , the bimodule property $\mathcal{L}$ gives a unique linear map

$$L_{kl} : \text{B}(\mathcal{R}_l, \mathcal{R}_k) \longrightarrow \text{B}(\mathcal{R}_l, \mathcal{R}_k) \quad (4.3)$$

such that

$$\mathcal{L}(A \otimes Z) = A \otimes L_{kl}(Z) \quad A \in \text{B}(\mathcal{K}_l, \mathcal{K}_k), \quad Z \in \text{B}(\mathcal{R}_l, \mathcal{R}_k). \quad (4.4)$$

Indeed, bimodule property restricts $\mathcal{L}(E_{ij}^{kl} \otimes Z) = E_{ij}^{kl} \otimes L_{kl}(Z)$ because of the multiplication by E_{i1}^{kk} on the left and E_{1j}^{ll} on the right, and hence for every $A \in \text{B}(\mathcal{K}_l, \mathcal{K}_k)$. This determines L_{kl} independently of the chosen bases. Since $\mathcal{L}(X^*) = \mathcal{L}(X)^*$,

$$L_{lk}(Z^*) = L_{kl}(Z)^*. \quad (4.5)$$

We call $\text{B}(\mathcal{R}_l, \mathcal{R}_k)$ a directed corner, a cross-corner when $k \neq l$, and a self-corner when $k = l$, and we write $S_t^{kl} := e^{tL_{kl}}$ as the restriction on the (k, l) corner. Since $\text{Ker } \mathcal{L} = \mathcal{N}$,

$$\text{Ker } L_{kl} = \{0\} \quad (k \neq l), \quad \text{Ker } L_{kk} = \mathbb{C}I_{\mathcal{R}_k}. \quad (4.6)$$

Moreover, $E\mathcal{L} = 0$ gives $\text{Tr}(\sigma_k L_{kk}(Z)) = 0$ for every $Z \in \text{B}(\mathcal{R}_k)$. Define

$$V_{kl} := \begin{cases} \text{B}(\mathcal{R}_l, \mathcal{R}_k), & k \neq l, \\ \{Z \in \text{B}(\mathcal{R}_k) : \text{Tr}(\sigma_k Z) = 0\}, & k = l. \end{cases} \quad (4.7)$$

We call V_{kl} the *centered corner space*. Each V_{kl} is invariant under S_t^{kl} .

For $Z, W \in \text{B}(\mathcal{R}_l, \mathcal{R}_k)$, set the corner KMS inner product as

$$h_{kl,1/2}(Z, W) := \text{Tr}(Z^* \sigma_k^{1/2} W \sigma_l^{1/2}). \quad (4.8)$$

For every faithful invariant state ω , the KMS inner product on $\mathbb{C}_{kl}$ factorizes as

$$\langle A \otimes Z, B \otimes W \rangle_{\text{KMS}, \omega} = \text{Tr}(\eta_l^{1/2} A^* \eta_k^{1/2} B) h_{kl,1/2}(Z, W).$$

The first factor is a positive definite inner product on $\mathcal{B}(\mathcal{K}_l, \mathcal{K}_k)$.

4.2. Weighted forms and endpoint interpolation. For $0 \leq s \leq 1$ and $Z, W \in \mathcal{B}(\mathcal{R}_l, \mathcal{R}_k)$, define

$$h_{kl,s}(Z, W) := \text{Tr}(Z^* \sigma_k^s W \sigma_l^{1-s}), \quad h_{kl,s}(Z) := h_{kl,s}(Z, Z), \quad (4.9)$$

$$q_{kl,s}(Z) := -\Re \text{Tr}(Z^* \sigma_k^s L_{kl}(Z) \sigma_l^{1-s}). \quad (4.10)$$

At $s = \frac{1}{2}$, $h_{kl,s}$ is the corner KMS inner product from Section 4.1. At the endpoints,

$$h_{kl,0}(Z) = \text{Tr}(Z^* Z \sigma_l), \quad q_{kl,0}(Z) = -\Re \text{Tr}(Z^* L_{kl}(Z) \sigma_l),$$

$$h_{kl,1}(Z) = \text{Tr}(Z Z^* \sigma_k), \quad q_{kl,1}(Z) = -\Re \text{Tr}(L_{kl}(Z) Z^* \sigma_k).$$

Kadison's inequality implies that S_t^{kl} is contractive

$$h_{kl,0}(S_t^{kl} Z) \leq h_{kl,0}(Z), \quad h_{kl,1}(S_t^{kl} Z) \leq h_{kl,1}(Z), \quad (4.11)$$

in both endpoint norms. Indeed, for $Y = E_{ij}^{kl} \otimes Z$,

$$P_t(Y)^* P_t(Y) \leq P_t(Y^* Y), \quad P_t(Y) P_t(Y)^* \leq P_t(Y Y^*).$$

Taking expectations in a faithful invariant state ω and cancelling the positive factors from η_l and η_k yields (4.11). Differentiation at $t = 0$ yields

$$q_{kl,0}(Z) \geq 0, \quad q_{kl,1}(Z) \geq 0. \quad (4.12)$$

Definition 4.1. For $a \in \{0, 1\}$, set

$$\kappa_{kl}^{(a)} := \inf_{0 \neq Z \in V_{kl}} \frac{q_{kl,a}(Z)}{h_{kl,a}(Z)}, \quad (4.13)$$

with value $+\infty$ when $V_{kl} = \{0\}$.

Fix a corner (k, l) and write $h_s = h_{kl,s}$, $q_s = q_{kl,s}$, $L = L_{kl}$, and $S_t = e^{tL}$.

Lemma 4.2. Suppose that $V_{kl} \neq \{0\}$, and let $\kappa_0, \kappa_1 \in \mathbb{R}$ satisfy

$$q_0(Z) \geq \kappa_0 h_0(Z), \quad q_1(Z) \geq \kappa_1 h_1(Z) \quad \text{for } Z \in V_{kl}. \quad (4.14)$$

Then, for every $s \in [0, 1]$ and $Z \in V_{kl}$,

$$q_s(Z) \geq ((1-s)\kappa_0 + s\kappa_1) h_s(Z). \quad (4.15)$$

Proof. Step 1: Weighted Hilbert spaces. Set $\mathbf{W}_{kl} := \mathcal{B}(\mathcal{R}_l, \mathcal{R}_k)$ and define

$$H_s := (\mathbf{W}_{kl}, h_s), \quad 0 \leq s \leq 1.$$

Since σ_k and σ_l are faithful, each h_s is an inner product. Relative to the Hilbert–Schmidt inner product, h_0 and h_1 are respectively given by the operator

$$G_0 Z = Z \sigma_l, \quad G_1 Z = \sigma_k Z. \quad (4.16)$$

These two operators are positive definite and commute. Note that the inner product h_s is given by the positive definite operator

$$G_s Z = \sigma_k^s Z \sigma_l^{1-s}$$

By simultaneously diagonalizing G_0 and G_1 and the weighted L_2 interpolation formula, we have the isometric identity of complex interpolation [4, Theorem 5.5.3].

$$[H_0, H_1]_s = H_s, \quad 0 < s < 1, \quad (4.17)$$

Step 2: Centering. If $k \neq l$, then $V_{kl} = W_{kl}$. If $k = l$, the map

$$\Pi : Z \mapsto Z - \text{Tr}(\sigma_k Z)I. \quad (4.18)$$

is the orthogonal projection onto V_{kk} in every H_s . The inclusion $V_{kk} \hookrightarrow W_{kk}$ and the projection Π are contractive at both endpoints. The operator interpolation theorem [4, Theorem 4.1.2] therefore gives the isometric identity

$$[\Pi H_0, \Pi H_1]_s = \Pi[H_0, H_1]_s = \Pi H_s, \quad 0 < s < 1. \quad (4.19)$$

Step 3: Endpoint interpolation. For $Z \in V_{kl}$ and $a = 0, 1$,

$$\frac{d}{dt} h_a(S_t Z) = 2\Re h_a(S_t Z, L S_t Z) = -2q_a(S_t Z).$$

Since V_{kl} is invariant under S_t , the endpoint bounds (4.14) and Grönwall's inequality yield

$$\|S_t|_{V_{kl}}\|_{H_a \rightarrow H_a} \leq e^{-\kappa_a t}, \quad a = 0, 1. \quad (4.20)$$

For $0 < s < 1$, interpolating $S_t|_{V_{kl}}$ between the endpoint spaces and using (4.17) and (4.19) gives

$$\|S_t|_{V_{kl}}\|_{H_s \rightarrow H_s} \leq \|S_t|_{V_{kl}}\|_{H_0 \rightarrow H_0}^{1-s} \|S_t|_{V_{kl}}\|_{H_1 \rightarrow H_1}^s \leq e^{-((1-s)\kappa_0 + s\kappa_1)t}. \quad (4.21)$$

Set $\kappa_s = (1-s)\kappa_0 + s\kappa_1$. Then

$$h_s(S_t Z) \leq e^{-2\kappa_s t} h_s(Z).$$

Differentiating at $t = 0^+$ gives

$$-2q_s(Z) = \left. \frac{d}{dt} h_s(S_t Z) \right|_{t=0^+} \leq -2\kappa_s h_s(Z),$$

which is (4.15). □

4.3. BKM decomposition and endpoint limits. Fix a faithful invariant state ω , and choose an orthonormal eigenbasis $\{|k, i\rangle\}_i$ for each noiseless block η_k . In these bases,

$$\omega = \bigoplus_k \eta_k \otimes \sigma_k, \quad \eta_k = \sum_i \lambda_{ki} |k, i\rangle \langle k, i|.$$

For $A = A^*$, set $X = \Gamma_\omega^{-1}(A)$. Then $X = X^*$ and $H_\omega(A) = \text{Tr}[X \Gamma_\omega(X)]$. Trace duality and Hilbert–Schmidt self-adjointness of Γ_ω give

$$Q_\omega(A) = -\text{Tr}[\mathcal{L}_*(\Gamma_\omega(X))X] = -\text{Tr}[\Gamma_\omega(X)\mathcal{L}(X)] = -\Re \text{Tr}[X^* \Gamma_\omega(\mathcal{L}(X))].$$

Consider a matrix-unit coordinate

$$Y = E_{ij}^{kl} \otimes Z, \quad Z \in \mathcal{B}(\mathcal{R}_l, \mathcal{R}_k),$$

and write

$$u = (k, i), \quad v = (l, j), \quad r = \frac{\lambda_{ki}}{\lambda_{lj}}.$$

If $u = v$, assume in addition that $Z = Z^*$. Set

$$X = \begin{cases} Y + Y^*, & u \neq v, \\ Y, & u = v, \end{cases} \quad A = \Gamma_\omega(X).$$

Lemma 4.3. *For the above choices, if $Z \neq 0$, then*

$$\frac{Q_\omega(A)}{H_\omega(A)} = R_{kl}(r, Z) := \frac{\int_0^1 r^s q_{kl,s}(Z) \, ds}{\int_0^1 r^s h_{kl,s}(Z) \, ds}. \quad (4.22)$$

Proof. By the choice of eigenbases,

$$\omega^s Y \omega^{1-s} = \lambda_{lj} r^s E_{ij}^{kl} \otimes \sigma_k^s Z \sigma_l^{1-s}. \quad (4.23)$$

Together with $\mathcal{L}(Y) = E_{ij}^{kl} \otimes L_{kl}(Z)$, this gives

$$\mathrm{Tr}[Y^* \Gamma_\omega(Y)] = \lambda_{lj} \int_0^1 r^s h_{kl,s}(Z) \, ds, \quad -\Re \mathrm{Tr}[Y^* \Gamma_\omega(\mathcal{L}(Y))] = \lambda_{lj} \int_0^1 r^s q_{kl,s}(Z) \, ds.$$

If $u \neq v$, then Y and Y^* lie in distinct matrix-unit coordinate spaces, which are orthogonal with respect to the Hilbert–Schmidt inner product. Both Γ_ω and $\mathcal{L}$ preserve these coordinates, so the cross terms vanish. Since both maps also preserve adjoints, the contributions from Y and Y^* are equal after taking real parts. Hence

$$H_\omega(A) = 2 \mathrm{Tr}[Y^* \Gamma_\omega(Y)], \quad Q_\omega(A) = -2\Re \mathrm{Tr}[Y^* \Gamma_\omega(\mathcal{L}(Y))].$$

If $u = v$, then $r = 1$ and $X = Y = Y^*$, so

$$H_\omega(A) = \mathrm{Tr}[Y^* \Gamma_\omega(Y)], \quad Q_\omega(A) = -\Re \mathrm{Tr}[Y^* \Gamma_\omega(\mathcal{L}(Y))].$$

Since $Z \neq 0$, we have $X \neq 0$ and hence $H_\omega(A) > 0$. Dividing the expressions for $Q_\omega(A)$ and $H_\omega(A)$ in either case proves (4.22). $\square$

For $r > 0$, set

$$J(r) := \int_0^1 r^s \, ds = \begin{cases} (r-1)/\log r, & r \neq 1, \\ 1, & r = 1, \end{cases} \quad \mu_r(ds) := \frac{r^s}{J(r)} \, ds. \quad (4.24)$$

For $Z \neq 0$, (4.22) can be written as

$$R_{kl}(r, Z) = \frac{\int_0^1 \frac{q_{kl,s}(Z)}{h_{kl,s}(Z)} h_{kl,s}(Z) \mu_r(ds)}{\int_0^1 h_{kl,s}(Z) \mu_r(ds)}.$$

Thus $R_{kl}(r, Z)$ is the weighted average of the quotients $q_{kl,s}(Z)/h_{kl,s}(Z)$ with respect to the measure $\mu_r(ds)$. Note that As $r \rightarrow 0^+$, $\mu_r \Rightarrow \delta_0$, and as $r \rightarrow \infty$, $\mu_r \Rightarrow \delta_1$ converges weakly. Indeed, write $r = e^{-x}$, where $x > 0$. For continuous $f \in C([0, 1])$, the substitution $t = xs$ gives

$$\int_0^1 f(s) \mu_r(ds) = \frac{1}{1 - e^{-x}} \int_0^x e^{-t} f(t/x) \, dt \longrightarrow f(0) \quad (x \rightarrow \infty).$$

The limit follows by dominated convergence. Replacing s with $1 - s$ gives the case $r \rightarrow \infty$.

Together with Lemma 4.2, these limits determine the infimum of R_{kl} .

Lemma 4.4. *For every fixed $0 \neq Z \in B(\mathcal{R}_l, \mathcal{R}_k)$,*

$$\lim_{r \rightarrow 0^+} R_{kl}(r, Z) = \frac{q_{kl,0}(Z)}{h_{kl,0}(Z)}, \quad \lim_{r \rightarrow \infty} R_{kl}(r, Z) = \frac{q_{kl,1}(Z)}{h_{kl,1}(Z)}. \quad (4.25)$$

Hence, for every directed corner,

$$\inf_{\substack{r > 0 \\ 0 \neq Z \in V_{kl}}} R_{kl}(r, Z) = \min \left\{ \kappa_{kl}^{(0)}, \kappa_{kl}^{(1)} \right\}. \quad (4.26)$$

Proof. Applying the weak convergence

$$\lim_{r \rightarrow 0^+} \mu_r = \delta_0, \quad \lim_{r \rightarrow \infty} \mu_r = \delta_1$$

limits to $s \mapsto q_{kl,s}(Z)$ and $s \mapsto h_{kl,s}(Z)$, proves the quotient limits. For the second assertion, Lemma 4.2 gives

$$q_{kl,s}(Z) \geq \min \{ \kappa_{kl}^{(0)}, \kappa_{kl}^{(1)} \} h_{kl,s}(Z) \quad \forall \quad 0 \leq s \leq 1, \quad Z \in V_{kl},$$

and integration proves the lower bound. Note that the limit (4.25) implies that every $0 \neq Z \in V_{kl}$,

$$\inf_{r > 0} R_{kl}(r, Z) \leq \frac{q_{kl,a}(Z)}{h_{kl,a}(Z)}, \quad a = 0, 1.$$

Taking the infimum over $Z \in V_{kl} \setminus \{0\}$ gives the reverse inequality. $\square$

4.4. Exact BKM coercivity. The ratios occurring in the BKM corner formula (4.22) are determined by the noiseless blocks η_k of the invariant state. A matrix-unit coordinate $Y = E_{ij}^{kl} \otimes Z$ carries the ratio

$$r = \frac{\lambda_{ki}}{\lambda_{lj}}.$$

As the infimum in Definition 2.10 ranges over all faithful invariant states, so the noiseless blocks η_k may vary. For a cross-corner ($k \neq l$), varying the relative block weights can realize every $r > 0$. For a self-corner $k = l$ with $\dim \mathcal{K}_k \geq 2$, the same holds for coordinates with $i \neq j$. We therefore define the set of *accessible corners*

$$\text{Acc} := \{(k, l) : k \neq l\} \cup \{(k, k) : \dim \mathcal{K}_k \geq 2\}, \quad (4.27)$$

and define endpoint constant

$$\kappa_{\partial} := \min_{\substack{(k,l) \in \text{Acc} \\ a \in \{0,1\}}} \kappa_{kl}^{(a)}. \quad (4.28)$$

Here the subscript ∂ refers to the two modular endpoints $s = 0, 1$. We interpret a minimum over an empty index set as $+\infty$.

On a self-corner with $\dim \mathcal{K}_k = 1$, we have $r = 1$, and define the *internal constant*

$$\gamma_k^{\text{int}} := \inf_{\substack{0 \neq Z = Z^* \\ \text{Tr}(\sigma_k Z) = 0}} \frac{\int_0^1 q_{kk,s}(Z) \, ds}{\int_0^1 h_{kk,s}(Z) \, ds}, \quad (4.29)$$

with value $+\infty$ when $V_{kk} = \{0\}$. This is the BKM coercivity constant for the primitive generator L_{kk} on $B(\mathcal{R}_k, \mathcal{R}_k)$.

Proposition 4.5. *Fix a faithful invariant state ω and the eigenbases from Section 4.3. Choose a total order on the composite indices $\{(k, i) : k \in \mathcal{I}, 1 \leq i \leq \dim \mathcal{K}_k\}$. For $A = A^*$, write*

$$X = \Gamma_\omega^{-1}(A) = \sum_{k,l,i,j} E_{ij}^{kl} \otimes Z_{ij}^{kl}.$$

Then

$$E_* A = 0 \iff EX = 0 \iff Z_{ij}^{kl} \in V_{kl} \text{ for all } k, l, i, j.$$

For $u = (k, i) \leq v = (l, j)$, define

$$X_{uv} := \begin{cases} E_{ij}^{kl} \otimes Z_{ij}^{kl} + E_{ji}^{lk} \otimes (Z_{ij}^{kl})^*, & u < v, \\ E_{ii}^{kk} \otimes Z_{ii}^{kk}, & u = v, \end{cases} \quad A_{uv} := \Gamma_\omega(X_{uv}).$$

Then $A = \sum_{u \leq v} A_{uv}$ is a self-adjoint decomposition, and $E_* A = 0$ implies $E_* A_{uv} = 0$ for every $u \leq v$. Moreover,

$$H_\omega(A) = \sum_{u \leq v} H_\omega(A_{uv}), \quad Q_\omega(A) = \sum_{u \leq v} Q_\omega(A_{uv}).$$

For $A \neq 0$, consequently,

$$\frac{Q_\omega(A)}{H_\omega(A)} = \sum_{\substack{u \leq v \\ A_{uv} \neq 0}} \frac{H_\omega(A_{uv})}{H_\omega(A)} \frac{Q_\omega(A_{uv})}{H_\omega(A_{uv})},$$

where the weights are positive and sum to one.

Proof. We first establish

$$E_* \Gamma_\omega = \Gamma_\omega E. \tag{4.30}$$

For $k \neq l$, both sides annihilate $E_{ij}^{kl} \otimes Z$. For $k = l$, equations (2.8)–(2.9) and $\text{Tr}(\sigma_k^s Z \sigma_k^{1-s}) = \text{Tr}(\sigma_k Z)$ give

$$\begin{aligned} E_* \Gamma_\omega(E_{ij}^{kk} \otimes Z) &= m(\lambda_{ki}, \lambda_{kj}) \text{Tr}(\sigma_k Z) E_{ij}^{kk} \otimes \sigma_k \\ &= \Gamma_\omega E(E_{ij}^{kk} \otimes Z), \end{aligned}$$

where m is the logarithmic mean. This proves (4.30). Since Γ_ω is invertible, $E_* A = 0$ is equivalent to $EX = 0$. By (2.8) and linear independence of the matrix units, $EX = 0$ holds if and only if $\text{Tr}(\sigma_k Z_{ij}^{kk}) = 0$ for all k, i, j , which is precisely the stated condition on V_{kl} .

Since Γ_ω^{-1} preserves adjoints, $X = X^*$ and $Z_{ji}^{lk} = (Z_{ij}^{kl})^*$. Thus the X_{uv} are self-adjoint and sum to X . Applying Γ_ω gives $A = \sum_{u \leq v} A_{uv}$ with $A_{uv} = A_{uv}^*$. If $EX = 0$, the conditions $\text{Tr}(\sigma_k Z_{ij}^{kk}) = 0$ also give $EX_{uv} = 0$ for every $u \leq v$. Equation (4.30) then yields $E_* A_{uv} = 0$.

Both Γ_ω and $\mathcal{L}$ preserve each subspace $E_{ij}^{kl} \otimes \text{B}(\mathcal{R}_l, \mathcal{R}_k)$. For $u < v$, the matrix units in X_{uv} correspond to (u, v) and (v, u) , whereas X_{uu} contains only the matrix unit corresponding to (u, u) . Different components therefore involve different matrix units. Their Hilbert–Schmidt orthogonality implies that all cross terms between distinct components vanish. Using

$\Gamma_\omega^{-1}(A_{uv}) = X_{uv}$, we obtain

$$\begin{aligned} H_\omega(A) &= \sum_{\substack{u \leq v \\ u' \leq v'}} \text{Tr}[X_{uv}^* \Gamma_\omega(X_{u'v'})] = \sum_{u \leq v} \text{Tr}[X_{uv}^* \Gamma_\omega(X_{uv})] = \sum_{u \leq v} H_\omega(A_{uv}), \\ Q_\omega(A) &= -\Re \sum_{\substack{u \leq v \\ u' \leq v'}} \text{Tr}[X_{uv}^* \Gamma_\omega(\mathcal{L}(X_{u'v'}))] = -\sum_{u \leq v} \Re \text{Tr}[X_{uv}^* \Gamma_\omega(\mathcal{L}(X_{uv}))] = \sum_{u \leq v} Q_\omega(A_{uv}). \end{aligned}$$

For $A \neq 0$, dividing by $H_\omega(A)$ gives the quotient formula. Positive definiteness of H_ω makes every nonzero-component weight positive, and additivity of H_ω shows that the weights sum to one. $\square$

With above proposition, we show that the BKM coercivity constant is characterized by the endpoint constant and the internal constant together .

Theorem 4.6. *Let P_t be a quantum Markov semigroup. Assume the limit condition expectation $E = \lim_{t \rightarrow \infty} P_t$ exists and admits the block decomposition as above. One has*

$$\lambda_{\text{BKM}} = \min \left\{ \kappa_\partial, \min_{k: \dim \mathcal{K}_k = 1} \gamma_k^{\text{int}} \right\}. \quad (4.31)$$

Proof. Set

$$m := \min \left\{ \kappa_\partial, \min_{k: \dim \mathcal{K}_k = 1} \gamma_k^{\text{int}} \right\}.$$

Fix a faithful invariant state ω and a nonzero admissible tangent vector A . By Proposition 4.5, it suffices to show that every nonzero component has quotient at least m . On an accessible corner,

$$R_{kl}(r, Z) \geq \min\{\kappa_{kl}^{(0)}, \kappa_{kl}^{(1)}\} \geq \kappa_\partial \geq m$$

by Lemma 4.4. On a self-corner with $\dim \mathcal{K}_k = 1$, $r = 1$ and $Z = Z^*$, so the quotient is at least $\gamma_k^{\text{int}} \geq m$. Hence $\lambda_{\text{BKM}} \geq m$.

For the reverse bound, fix k with $\dim \mathcal{K}_k = 1$ and $V_{kk} \neq \{0\}$. Choose $0 \neq Z = Z^* \in V_{kk}$ and set

$$X = E_{11}^{kk} \otimes Z, \quad A = \Gamma_\omega(X).$$

Proposition 4.5 gives $E_* A = 0$, and Lemma 4.3 identifies the quotient $Q_\omega(A)/H_\omega(A)$ with the ratio in (4.29). Taking the infimum over Z gives $\lambda_{\text{BKM}} \leq \gamma_k^{\text{int}}$.

Next fix $(k, l) \in \text{Acc}$, $r > 0$, and $0 \neq Z \in V_{kl}$. Choose distinct noiseless basis labels (k, i) and (l, j) . Assign weights r and 1 to the selected labels and weight 1 to every remaining label. Dividing all weights by their sum gives positive diagonal blocks η_u and a faithful invariant state

$$\omega = \bigoplus_u \eta_u \otimes \sigma_u, \quad \frac{\lambda_{ki}}{\lambda_{lj}} = r.$$

Let

$$Y := E_{ij}^{kl} \otimes Z, \quad X := Y + Y^*, \quad A := \Gamma_\omega(X).$$

Proposition 4.5 gives $E_* A = 0$, and Lemma 4.3 gives

$$\frac{Q_\omega(A)}{H_\omega(A)} = R_{kl}(r, Z).$$

Taking the infimum over r and Z and applying Lemma 4.4, we obtain

$$\lambda_{\text{BKM}} \leq \inf_{\substack{r > 0 \\ 0 \neq Z \in V_{kl}}} R_{kl}(r, Z) = \min\{\kappa_{kl}^{(0)}, \kappa_{kl}^{(1)}\}.$$

Minimizing over all accessible corners gives $\lambda_{\text{BKM}} \leq \kappa_{\partial}$, proving (4.31). $\square$

Under KMS symmetry, Corollary 4.7, the finiteness of $\mathcal{I}$, and the nonnegativity of the endpoint constants give (4.33).

Corollary 4.7. *Let $P_t = e^{t\mathcal{L}}$ be a quantum Markov semigroup that is KMS symmetric with respect to a faithful invariant state. Then*

$$\gamma_k^{\text{int}} > 0, \quad \text{if} \quad \dim \mathcal{K}_k = 1. \quad (4.32)$$

As a consequence,

$$\alpha(\mathcal{L}) > 0 \iff \lambda_{\text{BKM}}(\mathcal{L}) > 0 \iff \kappa_{\partial} > 0. \quad (4.33)$$

Proof. If $V_{kk} = \{0\}$, then $\gamma_k^{\text{int}} = +\infty$ by convention. Otherwise, (4.12) and the decay of $h_{kk,s}(Z)$ in Lemma 4.2 gives $q_{kk,s}(Z) \geq 0$ for every $s \in [0, 1]$ and $Z \in V_{kk}$. By KMS symmetry and Section 4.1, L_{kk} is self-adjoint for $h_{kk,1/2}$ and has kernel $\mathbb{C}I$. Since V_{kk} is the $h_{kk,1/2}$ -orthogonal complement of $\mathbb{C}I$, there is $\lambda_k > 0$ such that

$$q_{kk,1/2}(Z) \geq \lambda_k h_{kk,1/2}(Z) \quad (Z \in V_{kk}).$$

For every nonzero $Z \in V_{kk}$, the integrand is nonnegative and strictly positive at $s = \frac{1}{2}$. Continuity in s therefore gives

$$\int_0^1 q_{kk,s}(Z) \, ds > 0,$$

which implies the quotient

$$\frac{\int_0^1 q_{kk,s}(Z) \, ds}{\int_0^1 h_{kk,s}(Z) \, ds}, \quad (4.34)$$

is positive for every Z on the set

$$\left\{ Z = Z^* : \text{Tr}(\sigma_k Z) = 0, \int_0^1 h_{kk,s}(Z) \, ds = 1 \right\}.$$

Since this set is compact, the integral quotient attains a positive minimum. $\square$

5. THE COMPLETE BKM CONSTANT AND THE GNS SPECTRAL GAP

In this section, we identify the complete endpoint constant κ_{∂}^c with the spectral gap of the GNS symmetrization and prove that the BKM coercivity constant stabilizes at ancillary dimension two. These identities together give the CMLSI criterion. We then consider the GNS-symmetric case.

5.1. The GNS symmetrization and its corner formula. For a faithful invariant state ω , define the GNS-inner product and norm as

$$\langle X, Y \rangle_{2,\omega} := \text{Tr}(\omega X^* Y), \quad \|X\|_{2,\omega}^2 := \langle X, X \rangle_{2,\omega}. \quad (5.1)$$

We denote by $L_2(\omega)$ the Hilbert space associated with the above inner product. Note that the conditional expectation E is the orthogonal projection from $L_2(\omega)$ onto $\mathcal{N}$ for every faithful invariant state ω [13, Section 2.4]. Indeed, since E preserves ω and is $\mathcal{N}$ -bimodular, for $Y \in \mathcal{N}$,

$$\langle Y, X \rangle_{2,\omega} = \text{Tr}(\omega E(Y^* X)) = \text{Tr}(\omega Y^* E(X)) = \langle Y, E(X) \rangle_{2,\omega}.$$

Thus E is self-adjoint on $L_2(\omega)$, and $\text{B}(\mathcal{H}) = \mathcal{N} \oplus^{\perp_\omega} \text{Ker } E$.

Proposition 5.1. *The $L_2(\omega)$ -adjoint of $\mathcal{L}$ is independent of the choice of faithful invariant state ω . Denote this adjoint by $\mathcal{L}^\sharp$ and define the GNS symmetrization by*

$$\mathcal{L}_{\text{GNS}} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\sharp). \quad (5.2)$$

The operator $\mathcal{L}_{\text{GNS}}$ is self-adjoint and non-positive on every $L_2(\omega)$. The subspace $\text{Ker } E$ is invariant under $\mathcal{L}$, $\mathcal{L}^\sharp$, and $\mathcal{L}_{\text{GNS}}$.

Proof. To identify the adjoint of $\mathcal{L}$, write

$$\omega = \bigoplus_k \eta_k \otimes \sigma_k$$

in the fixed-point decomposition of Proposition 2.4. The directed corners $\mathcal{C}_{kl} = \text{B}(\mathcal{K}_l, \mathcal{K}_k) \otimes \text{B}(\mathcal{R}_l, \mathcal{R}_k)$ form an orthogonal direct sum in $L_2(\omega)$, and (4.4) gives $\mathcal{L}|_{\mathcal{C}_{kl}} = \text{id} \otimes L_{kl}$. For $A, B \in \text{B}(\mathcal{K}_l, \mathcal{K}_k)$ and $Z, W \in \text{B}(\mathcal{R}_l, \mathcal{R}_k)$,

$$\langle A \otimes Z, B \otimes W \rangle_{2,\omega} = \text{Tr}(\eta_l A^* B) h_{kl,0}(Z, W), \quad h_{kl,0}(Z, W) = \text{Tr}(Z^* W \sigma_l). \quad (5.3)$$

Hence the adjoint of $\mathcal{L}$ on this corner is $\text{id} \otimes L_{kl}^\sharp$, where $L_{kl}^\sharp$ is the adjoint of L_{kl} for $h_{kl,0}$. Each corner adjoint depends only on L_{kl} and $h_{kl,0}$, so the adjoint of $\mathcal{L}$ is independent of the noiseless blocks η_k .

Finally, Kadison's inequality and invariance of ω imply $\|P_t X\|_{2,\omega} \leq \|X\|_{2,\omega}$. Differentiating at zero gives

$$\langle X, -\mathcal{L}_{\text{GNS}} X \rangle_{2,\omega} = \frac{1}{2}(\langle X, -\mathcal{L} X \rangle_{2,\omega} + \langle -\mathcal{L} X, X \rangle_{2,\omega}) = -\Re \langle X, \mathcal{L} X \rangle_{2,\omega} \geq 0.$$

Since $E\mathcal{L} = \mathcal{L}E = 0$ and E is self-adjoint on $L_2(\omega)$, taking $L_2(\omega)$ -adjoints also gives $E\mathcal{L}^\sharp = \mathcal{L}^\sharp E = 0$. Hence $\text{Ker } E$ is invariant under $\mathcal{L}$, $\mathcal{L}^\sharp$, and $\mathcal{L}_{\text{GNS}}$. $\square$

We note that the operator $\mathcal{L}_{\text{GNS}}$ need not preserve adjoints and hence need not generate a QMS; see Section 6.3.

For a general QMS, extend the definition of the GNS spectral gap by

$$\lambda_{\text{GNS}}(\mathcal{L}) := \min \text{spec}(-\mathcal{L}_{\text{GNS}}|_{\text{Ker } E}) = \inf_{\substack{X \neq 0 \\ E(X)=0}} \frac{-\Re \text{Tr}(\omega X^* \mathcal{L}(X))}{\text{Tr}(\omega X^* X)}. \quad (5.4)$$

The infimum in (5.4) is independent of the chosen faithful invariant state ω . If $\mathcal{L}$ is GNS symmetric, then $\mathcal{L}^\sharp = \mathcal{L}$, and (5.4) recovers the definition in Section 2.1. In general, $\lambda_{\text{GNS}}(\mathcal{L})$ can vanish even in finite dimensions (Section 6.4).

Remark 5.2. The above definition agrees with the GNS spectral gap considered in [32, Remark 3.9]. In particular, $\lambda_{\text{GNS}}(\mathcal{L})$ is the largest constant λ such that

$$\|P_t X\|_{2,\omega} \leq e^{-\lambda t} \|X\|_{2,\omega}, \quad t \geq 0, \quad X \in \text{Ker } E.$$

Define the *complete endpoint constant* by

$$\kappa_{\partial}^c := \inf_{n \geq 1} \kappa_{\partial}(\text{id}_{M_n} \otimes \mathcal{L}) \quad (5.5)$$

Here for the matrix amplification $\mathcal{L} \otimes \text{id}_{M_n}$, $\mathcal{K}_k$ is replaced by $\mathbb{C}^n \otimes \mathcal{K}_k$, while σ_k , L_{kl} , and V_{kl} remain unchanged. For $n \geq 2$, every nonzero centered corner has two distinct noiseless coordinates, so its weight ratio r can vary over $(0, \infty)$, which leads the following amplification stabilization.

Proposition 5.3. *The GNS spectral gap satisfies*

$$\lambda_{\text{GNS}}(\mathcal{L}) = \min_{\substack{k,l \\ V_{kl} \neq \{0\}}} \kappa_{kl}^{(0)} = \kappa_{\partial}^c. \quad (5.6)$$

Consequently, for every $n \geq 1$,

$$\lambda_{\text{GNS}}(\text{id}_{M_n} \otimes \mathcal{L}) = \lambda_{\text{GNS}}(\mathcal{L}). \quad (5.7)$$

Proof. Choose an eigenbasis of each noiseless block $\eta_k = \sum_i \lambda_{ki} |k, i\rangle \langle k, i|$ and write

$$X = \sum_{k,l,i,j} E_{ij}^{kl} \otimes Z_{ij}^{kl}.$$

By (2.8), the condition $E(X) = 0$ is equivalent to $Z_{ij}^{kl} \in V_{kl}$ for every component. Orthogonality of the matrix-unit components and (4.4) give

$$\text{Tr}(\omega X^* X) = \sum_{k,l,i,j} \lambda_{lj} h_{kl,0}(Z_{ij}^{kl}), \quad (5.8)$$

$$-\Re \text{Tr}(\omega X^* \mathcal{L}(X)) = \sum_{k,l,i,j} \lambda_{lj} q_{kl,0}(Z_{ij}^{kl}). \quad (5.9)$$

Thus the global quotient is a weighted average of the corner quotients $q_{kl,0}/h_{kl,0}$, with strictly positive weights $\lambda_{lj} h_{kl,0}(Z_{ij}^{kl})$ for the nonzero components. Conversely, taking $X = E_{ij}^{kl} \otimes Z$ with $0 \neq Z \in V_{kl}$ gives the quotient $q_{kl,0}(Z)/h_{kl,0}(Z)$. This proves the first equality in (5.6).

Since $\mathcal{L}$ preserves adjoints, $L_{lk}(Z^*) = L_{kl}(Z)^*$, and hence

$$h_{kl,1}(Z) = h_{lk,0}(Z^*), \quad q_{kl,1}(Z) = q_{lk,0}(Z^*). \quad (5.10)$$

Since $Z \mapsto Z^*$ is a bijection from V_{kl} onto V_{lk} , we have $\kappa_{kl}^{(1)} = \kappa_{lk}^{(0)}$. Taking the minimum over k, l and using (5.5) gives the second equality in (5.6). Under amplification, $\mathcal{K}_k$ is replaced by $\mathbb{C}^n \otimes \mathcal{K}_k$, while σ_k , L_{kl} , and V_{kl} remain unchanged. This proves (5.7). $\square$

5.2. Complete BKM coercivity and the CMLSI criterion. We now show the complete BKM coercivity constant coincides with GNS spectral gap.

Theorem 5.4. *Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Then*

$$\lambda_{\text{BKM}}^c(\mathcal{L}) = \lambda_{\text{BKM}}(\text{id}_{M_2} \otimes \mathcal{L}) = \lambda_{\text{GNS}}(\mathcal{L}). \quad (5.11)$$

Proof. Note that for each amplification $\text{id}_{M_n} \otimes \mathcal{L}$ with $n \geq 2$, $\mathcal{K}_k$ is replaced by $\mathbb{C}^n \otimes \mathcal{K}_k$, while σ_k , L_{kl} , and V_{kl} remain unchanged. Hence the second term in the minimum formula of Theorem 4.6 always empty. Hence

$$\lambda_{\text{BKM}}(\text{id}_{M_n} \otimes \mathcal{L}) = \lambda_{\text{BKM}}(\text{id}_{M_2} \otimes \mathcal{L}) = \min_{\substack{k,l \\ V_{kl} \neq \{0\}}} \kappa_{kl}^{(0)} = \kappa_{\partial}^c.$$

which is exactly $\lambda_{\text{GNS}}(\mathcal{L})$ by Proposition 5.3. Testing with product states and product perturbations gives $\lambda_{\text{BKM}}(\text{id}_{M_2} \otimes \mathcal{L}) \leq \lambda_{\text{BKM}}(\mathcal{L})$. Taking the infimum over $n \geq 1$ completes the proof. $\square$

Substituting the above identity into Corollary 3.4, we obtain the following quantitative criterion for complete MLSI constant.

Corollary 5.5 (CMLSI criterion). *Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Then*

$$\frac{\lambda_{\text{GNS}}(\mathcal{L})}{2C_{\text{cb}}(E)} \leq \alpha^c(\mathcal{L}) \leq \lambda_{\text{GNS}}(\mathcal{L}). \quad (5.12)$$

Consequently,

$$\alpha^c(\mathcal{L}) > 0 \iff \lambda_{\text{GNS}}(\mathcal{L}) > 0. \quad (5.13)$$

Remark 5.6. The same bound was obtained in [13] for GNS-symmetric QMS. Corollary 5.5 extends this bound to QMS with a faithful asymptotic conditional expectation.

5.3. GNS-symmetric cases. Let $P_t = e^{t\mathcal{L}}$ be a quantum Markov semigroup that is GNS symmetric with respect to a faithful invariant state. It follows that $P_t = e^{t\mathcal{L}}$ is GNS symmetric with respect to every faithful invariant state [13]. Then $\mathcal{L}$ commutes with the modular group of every faithful invariant state ω , and also self-adjoint with respect to the inner product

$$\langle X, Y \rangle_{\omega, s} := \text{Tr}(X^* \omega^s Y \omega^{1-s}), \quad (5.14)$$

Consequently, all associated coercivity constants, including GNS ($s = 0$) and KMS ($s = 1/2$), are determined by the spectrum of $\mathcal{L}$,

$$\inf_{\substack{X \neq 0 \\ E(X)=0}} \frac{-\langle X, \mathcal{L}(X) \rangle_{\omega, s}}{\langle X, X \rangle_{\omega, s}} = \min(\text{spec}(-\mathcal{L}) \setminus \{0\}) =: \lambda \quad (5.15)$$

for $s \in [0, 1]$. This identifies the GNS and KMS gaps with λ . Note that $\mathcal{L}$ is also self-adjoint with respect to the BKM inner product as it is an average of s -inner product. It follows that all spectral gaps coincides with the minimum positive eigenvalue of $-\mathcal{L}$.

Corollary 5.7. *If $\mathcal{L}$ is GNS symmetric with respect to a faithful invariant state, then*

$$\lambda_{\text{KMS}}(\mathcal{L}) = \lambda_{\text{GNS}}(\mathcal{L}) = \lambda_{\text{BKM}}(\mathcal{L}) = \lambda_{\text{BKM}}^c(\mathcal{L}) = \min(\text{spec}(-\mathcal{L}) \setminus \{0\}). \quad (5.16)$$

6. EXAMPLES AND COUNTEREXAMPLES

This section gives two counterexamples separating the spectral and entropic quantities studied above. The primitive M_2 example has a positive MLSI constant but zero CMLSI constant, while the non-primitive M_3 example has a positive KMS spectral gap but zero BKM and MLSI constants. Both examples are KMS-symmetric. We begin with a Carré-du-Champ identity and a CMLSI criterion for primitive semigroups.

6.1. A CMLSI criterion for primitive semigroups. Let $P_t = e^{t\mathcal{L}}$ be a QMS with a faithful asymptotic conditional expectation E . Let σ be a faithful invariant state. For $Z \in \mathcal{B}(\mathcal{H})$, write

$$h_0(Z) := \|Z\|_{2,\sigma}^2 = \text{Tr}(\sigma Z^* Z), \quad q_0(Z) := -\Re \text{Tr}(\sigma Z^* \mathcal{L}(Z)). \quad (6.1)$$

Recall the Carré-du-Champ operator defined as

$$\Gamma_{\mathcal{L}}(Z) := \mathcal{L}(Z^* Z) - \mathcal{L}(Z^*) Z - Z^* \mathcal{L}(Z). \quad (6.2)$$

Suppose that the generator $\mathcal{L}$ admits the GKLS form of (2.1)

$$\mathcal{L}(X) = i[H, X] + \sum_j \left(V_j^* X V_j - \frac{1}{2} \{V_j^* V_j, X\} \right), \quad (6.3)$$

where V_j are the jump operators. Then, for every $Z \in \mathcal{B}(\mathcal{H})$,

$$\Gamma_{\mathcal{L}}(Z) = \sum_j [V_j, Z]^* [V_j, Z], \quad q_0(Z) = \frac{1}{2} \text{Tr}(\sigma \Gamma_{\mathcal{L}}(Z)) = \frac{1}{2} \sum_j \| [V_j, Z] \|_{2,\sigma}^2. \quad (6.4)$$

Define the *jump commutant* by

$$\mathcal{D}_+(\mathcal{L}) := \{Z : \Gamma_{\mathcal{L}}(Z) = 0\} = \{Z : [V_j, Z] = 0 \ \forall j\}. \quad (6.5)$$

For a primitive QMS with faithful invariant state σ , we have

$$E(X) = \text{Tr}(\sigma X)I, \quad E_*(\rho) = \text{Tr}(\rho)\sigma, \quad V_\sigma := \{Z \in \mathcal{B}(\mathcal{H}) : \text{Tr}(\sigma Z) = 0\}. \quad (6.6)$$

Corollary 6.1. *Let $\mathcal{L}$ generate a finite-dimensional primitive QMS that is KMS symmetric with respect to a faithful invariant state σ . Then $\alpha(\mathcal{L}) > 0$. Moreover,*

$$\alpha^c(\mathcal{L}) > 0 \iff \mathcal{D}_+(\mathcal{L}) = \mathbb{C}I. \quad (6.7)$$

Proof. The positivity of $\alpha(\mathcal{L})$ follows from Corollary 4.7 in the primitive case. Proposition 5.3 gives

$$\lambda_{\text{GNS}}(\mathcal{L}) = \kappa_{\partial}^c(\mathcal{L}) = \inf_{0 \neq Z \in V_\sigma} \frac{q_0(Z)}{\|Z\|_{2,\sigma}^2}. \quad (6.8)$$

If $\mathcal{D}_+(\mathcal{L}) = \mathbb{C}I$, the faithfulness of σ implies $q_0(Z) > 0$ for every nonzero $Z \in V_\sigma$. Compactness of the unit sphere in V_σ then gives the above infimum $\lambda_{\text{GNS}}(\mathcal{L}) > 0$. Conversely, if $Z \in \mathcal{D}_+(\mathcal{L}) \setminus \mathbb{C}I$, then

$$Z^\circ := Z - \text{Tr}(\sigma Z)I \in V_\sigma, \quad Z^\circ \neq 0, \quad q_0(Z^\circ) = 0.$$

Thus $\lambda_{\text{GNS}}(\mathcal{L}) = 0$. Corollary 5.5 then gives the stated CMLSI criterion. $\square$

6.2. Positive CMLSI without GNS symmetry on M_3 . Let

$$\sigma = \frac{1}{6} \text{diag}(1, 4, 1), \quad W_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad W_2 = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (6.9)$$

Set $K = W_1^* W_1 + W_2^* W_2 = \text{diag}(4, 2, 4)$ and define

$$\mathcal{L}(X) = W_1^* X W_1 + W_2^* X W_2 - \frac{1}{2} \{K, X\}. \quad (6.10)$$

Together with $[\sigma^{1/2}, K] = 0$, the identities

$$\sigma^{1/2} W_1^* \sigma^{-1/2} = W_2, \quad \sigma^{1/2} W_2^* \sigma^{-1/2} = W_1, \quad \mathcal{L}_*(\sigma) = 0 \quad (6.11)$$

show that $\mathcal{L}$ is KMS symmetric with respect to σ . For $X = (x_{ij})$, one has

$$\mathcal{L}(X) = \begin{pmatrix} 4(x_{22} - x_{11}) & 2x_{23} - 3x_{12} & -4x_{13} \\ 2x_{32} - 3x_{21} & x_{11} + x_{33} - 2x_{22} & 2x_{12} - 3x_{23} \\ -4x_{31} & 2x_{21} - 3x_{32} & 4(x_{22} - x_{33}) \end{pmatrix}. \quad (6.12)$$

Hence $\text{Ker } \mathcal{L} = \mathbb{C}I$, so the semigroup is primitive, with

$$E(X) = \text{Tr}(\sigma X)I, \quad E_*(\rho) = \text{Tr}(\rho)\sigma.$$

It is not GNS symmetric, since

$$\langle E_{12}, \mathcal{L}(E_{23}) \rangle_{\text{GNS}, \sigma} = \frac{4}{3}, \quad \langle \mathcal{L}(E_{12}), E_{23} \rangle_{\text{GNS}, \sigma} = \frac{1}{3}. \quad (6.13)$$

The self-adjoint part of V_σ has the real basis

$$\begin{aligned} Z_0 &= \text{diag}(1, 0, -1), & \mathcal{L}(Z_0) &= -4Z_0, \\ Z_1 &= \text{diag}(2, -1, 2), & \mathcal{L}(Z_1) &= -6Z_1, \\ Z_x^\pm &= E_{12} \pm E_{23} + E_{21} \pm E_{32}, & \mathcal{L}(Z_x^\pm) &= -(3 \mp 2)Z_x^\pm, \\ Z_y^\pm &= i(E_{12} \pm E_{23} - E_{21} \mp E_{32}), & \mathcal{L}(Z_y^\pm) &= -(3 \mp 2)Z_y^\pm, \\ Z_{13,x} &= E_{13} + E_{31}, & \mathcal{L}(Z_{13,x}) &= -4Z_{13,x}, \\ Z_{13,y} &= i(E_{13} - E_{31}), & \mathcal{L}(Z_{13,y}) &= -4Z_{13,y}. \end{aligned} \quad (6.14)$$

These basis vectors are BKM-orthogonal. Indeed, the diagonal directions Z_0, Z_1 are orthogonal, and the adjacent matrix units have the same BKM weight:

$$\Gamma_\sigma(E_{12}) = \frac{E_{12}}{2 \log 4}, \quad \Gamma_\sigma(E_{23}) = \frac{E_{23}}{2 \log 4}. \quad (6.15)$$

The same identities hold for their adjoints, while the remaining off-diagonal directions are orthogonal to them. Since the basis vectors are eigenvectors of $\mathcal{L}$, the integrated dissipation form is diagonal as well. Therefore,

$$\lambda_{\text{BKM}}(\mathcal{L}) = \lambda_{\text{KMS}}(\mathcal{L}) = \min\{4, 6, 1, 5, 4\} = 1. \quad (6.16)$$

The products

$$W_1 W_2 = 2 \text{diag}(1, 1, 0), \quad W_2 W_1 = 2 \text{diag}(0, 1, 1)$$

show that a matrix commuting with both W_1 and W_2 must be diagonal. Commutation with W_1 then forces its diagonal entries to be equal. Thus $\mathcal{D}_+(\mathcal{L}) = \mathbb{C}I$, and Corollary 6.1 gives $\alpha^c(\mathcal{L}) > 0$.

To compute the complete BKM constant, consider

$$\mathcal{W}_+ = \text{span}_{\mathbb{C}}\{E_{12}, E_{23}\}, \quad \mathcal{W}_- = \text{span}_{\mathbb{C}}\{E_{21}, E_{32}\}.$$

For $Z = xE_{12} + yE_{23} \in \mathcal{W}_+$, a direct calculation gives

$$h_0(Z) = \frac{4|x|^2 + |y|^2}{6}, \quad q_0(Z) = \frac{12|x|^2 + 3|y|^2 - 10\Re(\bar{x}y)}{6}. \quad (6.17)$$

Hence $q_0(Z) \geq \frac{1}{2}h_0(Z)$, with equality when $y = 2x$. On $\mathcal{W}_-$, the same formulas hold after interchanging x and y , so the minimum quotient is again $1/2$.

The spaces $\mathcal{W}_+$ and $\mathcal{W}_-$, the centered diagonal subspace, and $\text{span}_{\mathbb{C}}\{E_{13}, E_{31}\}$ form a GNS-orthogonal invariant decomposition of V_σ . On the latter two subspaces, the quotient

q_0/h_0 is at least 4. The global minimum is therefore $1/2$, attained at $W_1 = E_{12} + 2E_{23}$. Equation (6.8) gives

$$\lambda_{\text{GNS}}(\mathcal{L}) = \inf_{0 \neq Z \in V_\sigma} \frac{q_0(Z)}{h_0(Z)} = \frac{q_0(W_1)}{h_0(W_1)} = \frac{1}{2}. \quad (6.18)$$

Theorem 5.4 then yields

$$\lambda_{\text{BKM}}^c(\mathcal{L}) = \lambda_{\text{GNS}}(\mathcal{L}) = \frac{1}{2} < \lambda_{\text{BKM}}(\mathcal{L}) = 1. \quad (6.19)$$

Finally, Proposition 2.6 gives

$$C(E) = \|\sigma^{-1}\|_\infty = 6, \quad C_{\text{cb}}(E) = \text{Tr}(\sigma^{-1}) = \frac{27}{2}. \quad (6.20)$$

Corollary 3.4 therefore yields

$$\frac{1}{12} \leq \alpha(\mathcal{L}) \leq 1, \quad \frac{1}{54} \leq \alpha^c(\mathcal{L}) \leq \frac{1}{2}. \quad (6.21)$$

6.3. MLSI without CMLSI on M_2 . Let

$$\sigma = \frac{1}{5} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 1/\sqrt{2} \\ \sqrt{2} & 0 \end{pmatrix}, \quad K = W^*W = \text{diag}\left(2, \frac{1}{2}\right), \quad (6.22)$$

and define

$$\mathcal{L}_2(X) = W^*XW - \frac{1}{2}\{K, X\}. \quad (6.23)$$

The identities

$$\sigma^{\frac{1}{2}}W^*\sigma^{-\frac{1}{2}} = W, \quad [\sigma^{\frac{1}{2}}, K] = 0, \quad \mathcal{L}_{2*}(\sigma) = 0 \quad (6.24)$$

show that $\mathcal{L}_2$ is KMS symmetric with respect to σ . For $X = \begin{pmatrix} x & y \\ z & t \end{pmatrix}$,

$$\mathcal{L}_2(X) = \begin{pmatrix} 2(t-x) & z - \frac{5}{4}y \\ y - \frac{5}{4}z & \frac{1}{2}(x-t) \end{pmatrix}. \quad (6.25)$$

Hence $\text{Ker } \mathcal{L}_2 = \mathbb{C}I$, so the semigroup is primitive. It is not GNS symmetric, since

$$\langle E_{12}, \mathcal{L}_2(E_{21}) \rangle_{\text{GNS}, \sigma} = \frac{4}{5}, \quad \langle \mathcal{L}_2(E_{12}), E_{21} \rangle_{\text{GNS}, \sigma} = \frac{1}{5}. \quad (6.26)$$

Its GNS symmetrization does not preserve adjoints, since

$$\frac{1}{2}(\mathcal{L}_2 + \mathcal{L}_2^\sharp)(E_{12} + E_{21}) = -\frac{5}{8}E_{12} + \frac{5}{4}E_{21}.$$

The self-adjoint part of V_σ has the real basis

$$Z_0 = \text{diag}(1, -\frac{1}{4}), \quad \mathcal{L}_2(Z_0) = -\frac{5}{2}Z_0, \quad (6.27)$$

$$Z_x = E_{12} + E_{21}, \quad \mathcal{L}_2(Z_x) = -\frac{1}{4}Z_x, \quad (6.28)$$

$$Z_y = i(E_{12} - E_{21}), \quad \mathcal{L}_2(Z_y) = -\frac{9}{4}Z_y. \quad (6.29)$$

The vector Z_0 is BKM-orthogonal to Z_x and Z_y , while

$$h_s(Z_x, Z_y) = \frac{i}{5}(4^{1-s} - 4^s), \quad \int_0^1 h_s(Z_x, Z_y) ds = 0. \quad (6.30)$$

The BKM inner product is therefore diagonal in this basis. Since the basis vectors are eigenvectors of $\mathcal{L}_2$, the integrated dissipation form is diagonal as well. Hence

$$\lambda_{\text{BKM}}(\mathcal{L}_2) = \lambda_{\text{KMS}}(\mathcal{L}_2) = \min \left\{ \frac{5}{2}, \frac{1}{4}, \frac{9}{4} \right\} = \frac{1}{4}. \quad (6.31)$$

Proposition 2.6 gives $C(E) = \|\sigma^{-1}\|_\infty = 5$, so Corollary 3.4 yields

$$\frac{1}{40} \leq \alpha(\mathcal{L}_2) \leq \frac{1}{4}. \quad (6.32)$$

The operator W satisfies $\text{Tr}(\sigma W) = 0$. Since $[W, W] = 0$,

$$q_0(W) = 0, \quad h_0(W) = \text{Tr}(\sigma W^* W) = \frac{4}{5} > 0. \quad (6.33)$$

We have $W \in \mathcal{D}_+(\mathcal{L}_2)$ (non-normality of W gives $W^* \notin \mathcal{D}_+(\mathcal{L}_2)$), which by Corollary 6.1 implies $\lambda_{\text{GNS}}(\mathcal{L}_2) = 0$. Theorem 5.4 and Proposition 2.11 therefore give

$$\alpha(\text{id}_{M_2} \otimes \mathcal{L}_2) = \alpha^c(\mathcal{L}_2) = 0. \quad (6.34)$$

We now construct faithful states on $M_2 \otimes M_2$ whose MLSI quotients tend to zero. For $r > 0$, set

$$\eta_r = \frac{1}{1+r} \text{diag}(r, 1), \quad \omega_r = \eta_r \otimes \sigma, \quad (6.35)$$

and

$$Y = E_{12} \otimes W, \quad X = Y + Y^*, \quad A_r = \Gamma_{\omega_r}(X). \quad (6.36)$$

Then

$$A_r = A_r^* \neq 0, \quad \text{Tr}(A_r) = 0, \quad (\text{id}_{M_2} \otimes E_*)(A_r) = 0. \quad (6.37)$$

A direct calculation gives

$$h_s(W) = \frac{2}{5}(4^s + 4^{-s}), \quad q_s(W) = \frac{3}{10}(4^s - 4^{-s}), \quad (6.38)$$

and Lemma 4.3 yields

$$R_2(r) := \frac{Q_{\omega_r}(A_r)}{H_{\omega_r}(A_r)} = \frac{3}{4} \frac{J(4r) - J(r/4)}{J(4r) + J(r/4)}. \quad (6.39)$$

Here Q_{ω_r} is the dissipation form for $\text{id}_{M_2} \otimes \mathcal{L}_2$, and J is defined in (4.24). As $r \downarrow 0$,

$$R_2(r) = \frac{3 \log 4}{4 \log(1/r)} + O\left(\frac{1}{\log^2(1/r)}\right) \rightarrow 0. \quad (6.40)$$

For fixed $r > 0$, set

$$\rho_{r,\delta} := \omega_r + \delta A_r.$$

Since $\text{Tr}(A_r) = 0$ and $\omega_r > 0$, this is a faithful state for every sufficiently small nonzero real δ . Moreover, $(\text{id}_{M_2} \otimes E_*)(\rho_{r,\delta}) = \omega_r$. By (2.30) and (2.31),

$$\lim_{\delta \rightarrow 0} \frac{\text{EP}_{\text{id}_{M_2} \otimes \mathcal{L}_2}(\rho_{r,\delta})}{2D(\rho_{r,\delta} \| \omega_r)} = R_2(r).$$

Choose $r_m \rightarrow 0$. For each m , choose $\delta_m \neq 0$ sufficiently small that ρ_{r_m, δ_m} is faithful and its MLSI quotient differs from $R_2(r_m)$ by less than $1/m$. The MLSI quotients of ρ_{r_m, δ_m} then tend to zero.

Remark 6.2. In [12, Theorem 4.10], it was proved that GNS-symmetric quantum Markov semigroups satisfy

$$\alpha^c \geq \frac{1}{2t_{cb}(0.1)}, \quad (6.41)$$

where the complete-positivity return time is defined by

$$t_{cb}(\varepsilon) := \inf\{t > 0 : (1 - \varepsilon)E \leq_{cp} P_t \leq_{cp} (1 + \varepsilon)E\}, \quad 0 < \varepsilon < 1.$$

Here $\Phi \leq_{cp} \Psi$ means that $\Psi - \Phi$ is completely positive.

The above example shows that (6.41) does not extend to all KMS-symmetric semigroups, answering the corresponding question in [12, Problem 6.3] in the negative. Indeed, the Choi matrix of $E(X) = \text{Tr}(\sigma X)I_2$ is $\sigma^\top \otimes I_2 > 0$, and $P_t \rightarrow E$. For every $0 < \varepsilon < 1$, the Choi matrices of $P_t - (1 - \varepsilon)E$ and $(1 + \varepsilon)E - P_t$ are therefore positive for all sufficiently large t . Hence $t_{cb}(\varepsilon) < \infty$. Moreover, $\text{id} \not\leq_{cp} 1.1E$, so continuity at $t = 0$ gives $t_{cb}(0.1) > 0$. Thus the right-hand side of (6.41) is strictly positive, whereas $\alpha^c(\mathcal{L}_2) = 0$.

6.4. A positive KMS gap without MLSI on M_3 . Consider

$$W = \begin{pmatrix} 0 & 1/\sqrt{2} & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad K = W^*W = \text{diag}\left(2, \frac{1}{2}, 1\right), \quad (6.42)$$

and we define

$$\mathcal{L}_3(X) = W^*XW - \frac{1}{2}\{K, X\}. \quad (6.43)$$

This is obtained by adjoining the scalar block 1 to the jump operator in Section 6.3. For every $0 < \varepsilon < 1$, the faithful state

$$\sigma_\varepsilon = \text{diag}\left(\frac{\varepsilon}{5}, \frac{4\varepsilon}{5}, 1 - \varepsilon\right) \quad (6.44)$$

is invariant under $\mathcal{L}_3$. The identities $\sigma_\varepsilon^{1/2}W^* = W\sigma_\varepsilon^{1/2}$ and $[\sigma_\varepsilon^{1/2}, K] = 0$ show that $\mathcal{L}_3$ is KMS symmetric with respect to σ_ε .

The kernel is

$$\mathcal{N} = \text{Ker } \mathcal{L}_3 = \mathbb{C}P \oplus \mathbb{C}Q. \quad (6.45)$$

where $P = E_{11} + E_{22}$ and $Q = E_{33}$. Indeed, if $\mathcal{L}_3(X) = 0$, invariance of σ_ε and (6.4) imply $[W, X] = 0$. Hence $\mathcal{L}_3(X) = \frac{1}{2}[K, X]$, so also $[K, X] = 0$. Since K has distinct diagonal entries, X is diagonal and $[W, X] = 0$ forces $X_{11} = X_{22}$.

The asymptotic expectation and its trace predual are

$$E(X) = \left(\frac{1}{5}X_{11} + \frac{4}{5}X_{22}\right)P + X_{33}Q, \quad (6.46)$$

$$E_*(\rho) = \text{Tr}(P\rho) \left(\frac{1}{5}E_{11} + \frac{4}{5}E_{22}\right) + \text{Tr}(Q\rho)Q. \quad (6.47)$$

The corresponding reference states are $\sigma_P = \text{diag}(1/5, 4/5)$ and $\sigma_Q = 1$.

The four corners PM_3P , QM_3Q , PM_3Q , and QM_3P are invariant under $\mathcal{L}_3$. Its restriction to PM_3P is $\mathcal{L}_2$, and it vanishes on QM_3Q . The GNS inner product restricted to PM_3P is ε

times that of Section 6.3, so $\mathcal{L}_3$ is not GNS symmetric. On PM_3Q , the corner matrix is

$$L_{PQ} = \begin{pmatrix} -\frac{3}{2} & \sqrt{2} \\ \frac{1}{\sqrt{2}} & -\frac{3}{4} \end{pmatrix}, \quad (6.48)$$

with eigenvalues

$$\lambda_{\pm} = \frac{-9 \pm \sqrt{73}}{8}. \quad (6.49)$$

Since $\mathcal{L}_3$ preserves adjoints, its restriction to QM_3P has the same eigenvalues. KMS symmetry therefore gives

$$\lambda_{\text{KMS}}(\mathcal{L}_3) = \min \left\{ \frac{1}{4}, \frac{9 - \sqrt{73}}{8} \right\} = \frac{9 - \sqrt{73}}{8} > 0. \quad (6.50)$$

On the corner $Q \rightarrow P$, take $|u\rangle = 2^{-1/2}|1\rangle + |2\rangle \in P\mathbb{C}^3$. Since $W|u\rangle = |u\rangle$ and $\langle 3|W = \langle 3|$, the operator $|u\rangle\langle 3|$ commutes with W . A direct calculation gives

$$h_{PQ,0}(u) = \frac{3}{2}, \quad q_{PQ,0}(u) = 0, \quad (6.51)$$

while

$$h_{PQ,1}(u) = \frac{9}{10}, \quad q_{PQ,1}(u) = \frac{3}{20}. \quad (6.52)$$

More generally, for every $s \in [0, 1]$,

$$h_{PQ,s}(u) = \frac{1}{2} \left(\frac{1}{5} \right)^s + \left(\frac{4}{5} \right)^s, \quad q_{PQ,s}(u) = \frac{1}{4} \left[\left(\frac{4}{5} \right)^s - \left(\frac{1}{5} \right)^s \right].$$

Since $P \neq Q$, this corner is endpoint accessible. The vanishing of $q_{PQ,0}(u)$, together with Theorem 4.6, gives the BKM coercivity $\lambda_{\text{BKM}}(\mathcal{L}_3) = 0$ hence by Corollary 4.7 the MLSI constant $\alpha(\mathcal{L}_3) = 0$. We also provide direct witness of the $\alpha(\mathcal{L}_3) = 0$. Set

$$X = |u\rangle\langle 3| + |3\rangle\langle u|, \quad A_{\varepsilon} = \Gamma_{\sigma_{\varepsilon}}(X).$$

Then

$$A_{\varepsilon} = A_{\varepsilon}^* \neq 0, \quad E_* A_{\varepsilon} = 0, \quad \text{Tr}(A_{\varepsilon}) = 0.$$

With $r = \varepsilon/(1 - \varepsilon)$, Lemma 4.3 gives

$$R_{\varepsilon} := \frac{Q_{\sigma_{\varepsilon}}(A_{\varepsilon})}{H_{\sigma_{\varepsilon}}(A_{\varepsilon})} = \frac{J(4r/5) - J(r/5)}{4J(4r/5) + 2J(r/5)}. \quad (6.53)$$

Here $Q_{\sigma_{\varepsilon}}$ is the dissipation form for $\mathcal{L}_3$, and J is defined in (4.24). As $\varepsilon \rightarrow 0$,

$$R_{\varepsilon} = \frac{\log 2}{3 \log((1 - \varepsilon)/\varepsilon)} + O\left(\frac{1}{\log^2(1/\varepsilon)}\right) \rightarrow 0.$$

For fixed $0 < \varepsilon < 1$, set

$$\rho_{\varepsilon,\delta} := \sigma_{\varepsilon} + \delta A_{\varepsilon}.$$

This is a faithful state for every sufficiently small nonzero real δ , and $E_* \rho_{\varepsilon,\delta} = \sigma_{\varepsilon}$. By (2.30) and (2.31),

$$\lim_{\delta \rightarrow 0} \frac{\text{EP}_{\mathcal{L}_3}(\rho_{\varepsilon,\delta})}{2D(\rho_{\varepsilon,\delta} \| \sigma_{\varepsilon})} = R_{\varepsilon}.$$

Choose $\varepsilon_m \rightarrow 0$. For each m , choose $\delta_m > 0$ sufficiently small that $\rho_{\varepsilon_m, \delta_m}$ is faithful and its MLSI quotient differs from R_{ε_m} by less than $1/m$. The MLSI quotients of $\rho_{\varepsilon_m, \delta_m}$ tend to zero. Therefore

$$\lambda_{\text{BKM}}(\mathcal{L}_3) = \alpha(\mathcal{L}_3) = \alpha^c(\mathcal{L}_3) = 0. \quad (6.54)$$

7. APPLICATION TO THE CKG GIBBS SAMPLER

We apply the CMLSI criterion to the Gibbs sampler introduced by Chen, Kastoryano, and Gilyén [7]. This sampler is KMS symmetric with respect to the Gibbs state. We show that the kernel of its GNS Dirichlet form coincides with its fixed-point algebra, which implies a positive CMLSI constant by Corollary 5.5.

7.1. The CKG sampler. Let $H = H^* \in \mathcal{B}(\mathcal{H})$ be a Hamiltonian and let $\beta > 0$ be inverse temperature. The Gibbs state is

$$\sigma_\beta = \frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})}. \quad (7.1)$$

Let $\{A_a : a \in \mathbf{A}\}$ be a finite family of operators satisfying $A_{a^*} = A_a^*$ for an involution $a \mapsto a^*$ in $\mathbf{A}$. Write $A_a(t) = e^{itH} A_a e^{-itH}$.

Fix $\sigma_E > 0$ and $\omega_\gamma > \beta\sigma_E^2/2$. Following [7, Eqs. (1.3)–(1.4)], define the filtered operators

$$\hat{A}_a(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) A_a(t) e^{-i\omega t} dt, \quad f(t) = \left(\sigma_E \sqrt{\frac{2}{\pi}} \right)^{1/2} e^{-\sigma_E^2 t^2}. \quad (7.2)$$

The Gaussian transition weight is

$$\gamma(\omega) = \exp\left[-\frac{(\omega + \omega_\gamma)^2}{2\sigma_\gamma^2}\right], \quad \sigma_\gamma^2 = \frac{2\omega_\gamma}{\beta} - \sigma_E^2 > 0. \quad (7.3)$$

The Lindbladian in Heisenberg picture is

$$\mathcal{L}(X) = i[B, X] + \sum_a \int_{\mathbb{R}} \gamma(\omega) \left(\hat{A}_a(\omega)^* X \hat{A}_a(\omega) - \frac{1}{2} \{ \hat{A}_a(\omega)^* \hat{A}_a(\omega), X \} \right) d\omega, \quad (7.4)$$

where $B = B^*$ is the Hamiltonian correction from [7, Lemma II.1], given in (7.12). The semigroup $P_t = e^{t\mathcal{L}}$, is KMS symmetric with respect to σ_β [7, Corollary II.2]. Let E denote its asymptotic conditional expectation; see Section 4.1.

7.2. The commutants of filtered jump operators. For $Z \in \mathcal{B}(\mathcal{H})$, define the Dirichlet form

$$q_0(Z) := -\Re \text{Tr}(\sigma_\beta Z^* \mathcal{L}(Z)).$$

Proposition 7.1. *For every $Z \in \mathcal{B}(\mathcal{H})$,*

$$q_0(Z) = \frac{1}{2} \sum_a \int_{\mathbb{R}} \gamma(\omega) \left\| [\hat{A}_a(\omega), Z] \right\|_{2, \sigma_\beta}^2 d\omega. \quad (7.5)$$

Consequently,

$$q_0(Z) = 0 \iff [\hat{A}_a(\omega), Z] = 0 \text{ for every } a \text{ and a.e. } \omega. \quad (7.6)$$

Proof. The Hamiltonian term cancels in $\Gamma_{\mathcal{L}}$, and invariance of σ_β gives $q_0(Z) = \frac{1}{2} \text{Tr}(\sigma_\beta \Gamma_{\mathcal{L}}(Z))$. Applying (6.4) with $V_{a,\omega} = \sqrt{\gamma(\omega)} \hat{A}_a(\omega)$, with summation over a and integration over ω , yields (7.5). The equivalence (7.6) follows from nonnegativity of the integrand, strict positivity of γ , and faithfulness of σ_β . $\square$

Let $\mathcal{B}_H := \{E' - E : E, E' \in \text{spec}(H)\}$ be the Bohr spectrum of H . Write

$$H = \sum_E E P_E, \quad A_{a,\nu} := \sum_{E' - E = \nu} P_{E'} A_a P_E, \quad A_a(t) = \sum_{\nu \in \mathcal{B}_H} e^{it\nu} A_{a,\nu}. \quad (7.7)$$

Lemma 7.2. *The kernel of q_0 is given by*

$$\begin{aligned} \text{Ker } q_0 &= \{\hat{A}_a(\omega) : a \in \mathbf{A}, \omega \in \mathbb{R}\}' \\ &= \{A_a(t) : a \in \mathbf{A}, t \in \mathbb{R}\}' \\ &= \{A_{a,\nu} : a \in \mathbf{A}, \nu \in \mathcal{B}_H\}'. \end{aligned} \quad (7.8)$$

Proof. Dominated convergence shows that each $\hat{A}_a$ is norm continuous. Thus commutation for a.e. ω implies commutation for every ω , and Proposition 7.1 gives the first equality.

Suppose that Z commutes with every $\hat{A}_a(\omega)$. For each a , the function $F_a(t) := f(t)[A_a(t), Z]$ belongs to $L_1(\mathbb{R}; \mathcal{B}(\mathcal{H}))$ and satisfies

$$\hat{F}_a(\omega) = [\hat{A}_a(\omega), Z] = 0.$$

The Fourier uniqueness theorem, applied entrywise, gives $F_a = 0$ a.e. [9, Theorem 4.33]. Since $f \neq 0$ a.e., $[A_a(t), Z] = 0$ a.e. Norm continuity then gives this equality for every t . The converse follows directly from (7.2), proving the second equality.

For the last equality, the Bohr decomposition (7.7) shows that commutation with every $A_{a,\nu}$ implies commutation with every $A_a(t)$. Conversely,

$$A_{a,\nu} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-it\nu} A_a(t) dt, \quad (7.9)$$

where the limit holds in norm. Hence commutation with every $A_a(t)$ implies commutation with each $A_{a,\nu}$. $\square$

7.3. Fixed-point algebra and CMLSI. Let

$$\mathfrak{A}_{\text{orb}} := C^*(A_a(t) : a \in \mathbf{A}, t \in \mathbb{R}). \quad (7.10)$$

be the C^* -subalgebra generated $A_a(t), t \in \mathbb{R}$. Set

$$R_{\text{diss}} := \sum_a \int_{\mathbb{R}} \gamma(\omega) \hat{A}_a(\omega)^* \hat{A}_a(\omega) d\omega, \quad R_{\text{diss},\nu} := \sum_{E' - E = \nu} P_{E'} R_{\text{diss}} P_E. \quad (7.11)$$

The Hamiltonian correction from [7, Lemma II.1] is given by function calculus

$$B = \frac{i}{2} \sum_{\nu \in \mathcal{B}_H} \tanh\left(\frac{\beta\nu}{4}\right) R_{\text{diss},\nu}. \quad (7.12)$$

Proposition 7.3. *The fixed-point algebra $\mathcal{N} := \text{Ker } \mathcal{L} = \text{Fix}(P_t) = \text{Ran } E$ satisfies*

$$\mathcal{N} = \text{Ker } q_0 = \mathfrak{A}'_{\text{orb}}. \quad (7.13)$$

The MLSI constants satisfy

$$\alpha(\mathcal{L}) \geq \alpha^c(\mathcal{L}) \geq \frac{\lambda_{\text{GNS}}(\mathcal{L})}{2C_{\text{cb}}(E)} > 0. \quad (7.14)$$

Proof. Each $\widehat{A}_a(\omega)$ belongs to $\mathfrak{A}_{\text{orb}}$, hence so does R_{diss} . The algebra $\mathfrak{A}_{\text{orb}}$ is invariant under modular automorphism group $\alpha_t(\cdot) = e^{itH} \cdot e^{-itH}$, since $\alpha_t(A_a(s)) = A_a(t+s)$. Thus, for every $\nu \in \mathcal{B}_H$,

$$R_{\text{diss},\nu} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-it\nu} \alpha_t(R_{\text{diss}}) dt \in \mathfrak{A}_{\text{orb}}, \quad (7.15)$$

where the limit holds in norm. Equation (7.12) therefore gives $B \in \mathfrak{A}_{\text{orb}}$.

Since $A_a(t)^* = A_{a^*}(t)$, Lemma 7.2 gives $\text{Ker } q_0 = \mathfrak{A}'_{\text{orb}}$. Every Z in this commutant commutes with B and all filtered jump operators and their adjoints. Hence $\mathcal{L}(Z) = 0$ by (7.4). Conversely, $\mathcal{L}(Z) = 0$ implies $q_0(Z) = 0$, proving (7.13).

Since $\text{Ker } q_0 = \mathcal{N}$ and $\mathcal{N} \cap \text{Ker } E = \{0\}$, (5.4) and compactness give

$$\lambda_{\text{GNS}}(\mathcal{L}) = \min_{\substack{E(Z)=0 \\ \|Z\|_{2,\sigma_\beta}=1}} q_0(Z) > 0.$$

Corollary 5.5 and $\alpha^c(\mathcal{L}) \leq \alpha(\mathcal{L})$ now give (7.14). $\square$

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